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Wireless communication system

Abstract

When a vehicle and a smartphone are in a second communication state, the vehicle can operate, and then if the vehicle and the smartphone are not in the second communication state for a predetermined time or longer, and if a main switch of the vehicle detects a predetermined operation, an emergency advertising is transmitted from the vehicle. Thereby, the smartphone can be used as a key that can receive the emergency information from the vehicle. By enabling the smartphone to be used as a key for the vehicle, it is possible to popularize a smartphone that can be used as a key for the vehicle.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of International Application No. PCT/JP2021/021738 filed on Jun. 8, 2021, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD
(1) The present disclosure relates to a wireless communication system suitable for use in a two-wheeled vehicle.
BACKGROUND ART
(2) In recent years, a theft detection system that detects a theft of a vehicle has been developed. For example, if a theft detection system described in Patent Literature 1 determines that a vehicle that cannot communicate well with another vehicle equipped with a communication terminal may be a stolen vehicle, the theft detection system acquires location information of the vehicle and retains the location information in a server, thereby enabling tracking of the stolen vehicle.
(3) A small electronic tag (transmitter) is known which uses Bluetooth (registered trademark), a short-range wireless communication standard, to link with a dedicated smartphone application to detect location information of belongings and prevent loss (see, for example, Non Patent Literature 1).
CITATION LIST
Patent Literature
(4) Patent Literature 1: JP4161816B
Non-Patent Literature

(5) Non Patent Literature 1: “Seekit (registered trademark)”, [online], Panasonic Corporation of India, [Sep. 18, 2019], Internet <URL: https://www.seekit.panasonic.com/>

SUMMARY OF INVENTION

(6) However, in a theft detection system that detects a stolen vehicle using wireless communication, gives location information, and retains the location information in a server to enable tracking of the stolen vehicle, if a communication method with a long communication distance, such as Bluetooth LE communication is used, there is a problem that it is difficult to actually find the stolen vehicle because an area where the stolen vehicle is assumed to be present is widened based on only the location information.

(7) An object of the present disclosure is to provide a wireless communication system that can easily find a stolen vehicle even when a communication method with a long communication distance, such as Bluetooth LE communication, is used to detect the stolen vehicle.

(8) A wireless communication system according to the present disclosure includes a vehicle including an operation unit and a first wireless communication circuit conforming to a first wireless method; a first wireless communication terminal which includes a second wireless communication circuit conforming to the first wireless method and available to communicate with the first wireless communication circuit; and a second wireless communication terminal which includes a third wireless communication circuit conforming to the first wireless method and available to communicate with the first wireless communication circuit, and a fourth wireless communication circuit conforming to a second wireless method different from the first wireless method. When the first wireless communication circuit of the vehicle and the second wireless communication circuit of the first wireless communication terminal are in a first communication state, the vehicle becomes operable. When the first wireless communication circuit of the vehicle and the third wireless communication circuit of the second wireless communication terminal are in a second communication state, the vehicle becomes operable. After the first wireless communication circuit of the vehicle and the third wireless communication circuit of the second wireless communication terminal are in the second communication state, if the first wireless communication circuit and the third wireless communication circuit do not enter into the second communication state for a predetermined time or longer, and if the operation unit of the vehicle detects a predetermined operation, the first wireless communication circuit of the vehicle transmits a packet including emergency information.

(9) According to the present disclosure, when the first wireless communication circuit of the vehicle and the third wireless communication circuit of the second wireless communication terminal are in a second communication state, the vehicle becomes operable. Further, after the first wireless communication circuit of the vehicle and the third wireless communication circuit of the second wireless communication terminal are in the second communication state, if the first wireless communication circuit and the third wireless communication circuit do not enter into the second communication state for a predetermined time or longer, and if the operation unit of the vehicle detects a predetermined operation, the first wireless communication circuit of the vehicle transmits a packet including emergency information. Thus, the second wireless communication terminal can be used as a key that can receive the emergency information from the vehicle. By enabling the second wireless communication terminal to be used as a key for the vehicle, it is possible to popularize a smartphone that can be used as a key for the vehicle. Even if the first wireless communication terminal is lost or stolen, it is possible to use the second wireless communication terminal as a key.

(10) In the wireless communication system according to the above structure, even when the first wireless communication circuit of the vehicle and the second wireless communication circuit of the first wireless communication terminal are in the first communication state, after the first wireless communication circuit of the vehicle and the third wireless communication circuit of the second wireless communication terminal enter into the second communication state, if the first wireless

communication circuit and the third wireless communication circuit do not enter into the second communication state for the predetermined time or longer, and if the operation unit of the vehicle detects the predetermined operation, the first wireless communication circuit of the vehicle transmits the packet including the emergency information.

(11) According to the present disclosure, even if the first wireless communication circuit of the vehicle and the second wireless communication circuit of the first wireless communication terminal are in the first communication state, and the vehicle is operable in the first communication state, when an abnormality occurs in the vehicle, the second wireless communication terminal can receive a packet containing emergency information transmitted from the first wireless communication circuit of the vehicle. Thereby, if the first wireless communication terminal is lost or stolen, the theft can be detected.

(12) In the wireless communication system according to the above structure, the first wireless communication terminal does not include a wireless communication circuit conforming to the second wireless method.

(13) According to the present disclosure, it is possible to use for a conventional electronic key as the first wireless communication terminal.

(14) In the wireless communication system according to the above structure, the second wireless communication terminal is configured to stop at least the third wireless communication circuit from entering into the second communication state via the fourth wireless communication circuit.

(15) According to the present disclosure, it is possible to disable the operation of the vehicle by using the second wireless communication terminal.

(16) In the wireless communication system according to the above structure, the first wireless method is Bluetooth (registered trademark).

(17) According to the present disclosure, since Bluetooth is used for the communication between the first wireless communication circuit of the vehicle and the third wireless communication circuit of the second wireless communication terminal, it is possible to notify the second wireless communication terminal of an abnormality in the vehicle without using a telematics service in the vehicle. Further, since there is no need to have a communication line or GPS function to use the telematics service, costs can be reduced.

(18) In the wireless communication system according to the above structure, the second wireless method is cellular.

(19) According to the present disclosure, when the second wireless communication terminal is within the range where cellular communication is possible, it is possible to receive emergency information from the vehicle via cellular communication.

(20) In the wireless communication system according to the above structure, the second wireless communication terminal has at least one authentication function of password authentication, fingerprint authentication, and face authentication.

(21) According to the present disclosure, it is possible to determine whether or not the person having the wireless commutation terminal is the authorized owner.

(22) In the wireless communication system according to the above structure, in the authentication function of the second wireless communication terminal, if the authentication is successful, at least the third wireless communication circuit continues to be in the second communication state, and if the authentication fails, at least the third wireless communication circuit stops entering into the second communication state.

(23) According to the present disclosure, use of the second wireless communication terminal can be prohibited to anyone other than the authorized owner.

(24) In the wireless communication system according to the above structure, at least the second wireless communication terminal is registerable in the vehicle.

(25) According to the present disclosure, since the second wireless communication terminal can be registered in the vehicle, it is possible to prohibit the use of wireless communication terminals

other than the registered second wireless communication terminal.
(26) In the wireless communication system according to the above structure, the first communication state and the second communication state are the same.
(27) According to the present disclosure, since the communication state between the first wireless commutation circuit of the vehicle and the second wireless communication circuit of the first wireless communication terminal is the same as the communication state between the first wireless communication circuit of the vehicle and the third wireless communication circuit of the second wireless communication terminal, processing can be simplified.
(28) A wireless communication device mountable on a vehicle having an operation unit according to the present disclosure includes a first wireless communication circuit conforming to a first wireless method. The wireless communication device is able to be communicated with a first wireless communication terminal, the first wireless communication terminal including a second wireless communication circuit conforming to the first wireless method and available to communicate with the first wireless communication circuit, the wireless communication device is able to be communicated with a second wireless communication terminal, the second wireless communication terminal including a third wireless communication circuit conforming to the first wireless method and available to communicate with the first wireless communication circuit, and a fourth wireless communication circuit conforming to a second wireless method different from the first wireless method. When the first wireless communication circuit and the second wireless communication circuit of the first wireless communication terminal are in a first communication state, the vehicle becomes operable. When the first wireless communication circuit and the third wireless communication circuit of the second wireless communication terminal are in a second communication state, the vehicle becomes operable. After the first wireless communication circuit and the third wireless communication circuit of the second wireless communication terminal are in the second communication state, if the first wireless communication circuit and the third wireless communication circuit do not enter into the second communication state for a predetermined time or longer, and if the operation unit of the vehicle detects a predetermined operation, the first wireless communication circuit transmits a packet including emergency information.
(29) According to the present disclosure, when the first wireless communication circuit and the third wireless communication circuit of the second wireless communication terminal are in a second communication state, the vehicle becomes operable. After the first wireless communication circuit and the third wireless communication circuit of the second wireless communication terminal are in the second communication state, if the first wireless communication circuit and the third wireless communication circuit do not enter into the second communication state for a predetermined time or longer, and if the operation unit of the vehicle detects a predetermined operation, the first wireless communication circuit transmits a packet including emergency information. Therefore, it is possible to use the second wireless communication terminal as a key that can receive emergency information from the vehicle. Further, by using a smartphone as the second wireless communication terminal, it is possible to popularize a smartphone that can be used as a key for the vehicle.
(30) In the wireless communication device according to the above structure, even when the first wireless communication circuit and the second wireless communication circuit of the first wireless communication terminal are in the first communication state, after the first wireless communication circuit and the third wireless communication circuit of the second wireless communication terminal enter into the second communication state, if the first wireless communication circuit and the third wireless communication circuit do not enter into the second communication state for the predetermined time or longer, and if the operation unit of the vehicle detects the predetermined operation, the first wireless communication circuit transmits the packet including the emergency information.
(31) According to the present disclosure, even if the first wireless communication circuit of the

vehicle and the second wireless communication circuit of the first wireless communication terminal are in the first communication state, and the vehicle is operable in the first communication state, when an abnormality occurs in the vehicle, the second wireless communication terminal can receive a packet containing emergency information transmitted from the first wireless communication circuit of the vehicle. Thereby, if the first wireless communication terminal is lost or stolen, the theft can be detected.
(32) In the wireless communication device according to the above structure, the first wireless communication terminal does not include a wireless communication circuit conforming to the second wireless method.
(33) According to the present disclosure, it is possible to use for a conventional electronic key as the first wireless communication terminal.
(34) In the wireless communication device according to the above structure, the second wireless communication terminal is configured to stop at least the third wireless communication circuit from entering into the second communication state via the fourth wireless communication circuit.
(35) According to the present disclosure, it is possible to disable the operation of the vehicle by using the second wireless communication terminal.
(36) In the wireless communication device according to the above structure, the first wireless method is Bluetooth (registered trademark).
(37) According to the present disclosure, Bluetooth is used for the communication between the first wireless communication circuit of the vehicle and the third wireless communication circuit of the second wireless communication terminal. Therefore, it is possible to notify the second wireless communication terminal of an abnormality in the vehicle without using a telematics service in the vehicle. Further, since there is no need to have a communication line or GPS function to use the telematics service, costs can be reduced.
(38) In the wireless communication device according to the above structure, the second wireless method is cellular.
(39) According to the present disclosure, when the second wireless communication terminal is within the range where cellular communication is possible, it is possible to receive emergency information from the vehicle via cellular communication.
(40) In the wireless communication device according to the above structure, the second wireless communication terminal has at least one authentication function of password authentication, fingerprint authentication, and face authentication.
(41) According to the present disclosure, it is possible to determine whether or not the person having the wireless commutation terminal is the authorized owner.
(42) In the wireless communication device according to the above structure, in the authentication function of the second wireless communication terminal, if the authentication is successful, at least the third wireless communication circuit continues to be in the second communication state, and if the authentication fails, at least the third wireless communication circuit stops entering into the second communication state.
(43) According to the present disclosure, use of the second wireless communication terminal can be prohibited to anyone other than the authorized owner.
(44) In the wireless communication device according to the above structure, at least the second wireless communication terminal is registerable in the vehicle.
(45) According to the present disclosure, since the second wireless communication terminal can be registered in the vehicle, it is possible to prohibit the use of wireless communication terminals other than the registered second wireless communication terminal.
(46) In the wireless communication device according to the above structure, the first communication state and the second communication state are the same.
(47) According to the present disclosure, since the communication state between the first wireless commutation circuit of the vehicle and the second wireless communication circuit of the first

wireless communication terminal is the same as the communication state between the first wireless communication circuit of the vehicle and the third wireless communication circuit of the second wireless communication terminal, processing can be simplified.
(48) According to the present disclosure, expensive parts such as acceleration sensors are not required, and security can be improved while allowing the vehicle to be operated using just a smartphone.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a diagram showing an aspect of using a wireless communication system according to a first embodiment;
(2) FIG. **2** is a block diagram showing a schematic configuration of a part of a vehicle and a schematic configuration of a vehicle ECU of the wireless communication system according to the first embodiment;
(3) FIG. **3** is a diagram showing output waveforms of a first advertising packet and a second advertising packet transmitted from the vehicle ECU of the wireless communication system according to the first embodiment;
(4) FIG. **4** is a diagram showing output waveforms of the first advertising packet and the second advertising packet transmitted from the vehicle ECU of the wireless communication system according to the first embodiment;
(5) FIG. **5** is a block diagram showing a schematic configuration of an information sharing user terminal of the wireless communication system according to the first embodiment;
(6) FIG. **6** is a block diagram showing a schematic configuration of a bicycle parking lot Bluetooth unit of the wireless communication system according to the first embodiment;
(7) FIG. **7** is a block diagram showing a schematic configuration of a server of the wireless communication system according to the first embodiment;
(8) FIG. **8** is a diagram showing a packet structure of LE Uncoded PHYs of an advertising packet of Bluetooth used in the wireless communication system according to the first embodiment;
(9) FIG. **9** is a diagram showing a data structure of AD Type: 0xFF of the advertising packet of Bluetooth used in the wireless communication system according to the first embodiment;
(10) FIG. **10** is a diagram showing an example of server notification information notified to a server device of the wireless communication system according to the first embodiment;
(11) FIG. **11** is a diagram showing an example of user notification information notified to a vehicle user terminal of the wireless communication system according to the first embodiment;
(12) FIG. **12** is a diagram showing an overview of an operation of the vehicle ECU of the wireless communication system according to the first embodiment;
(13) FIG. **13** is a sequence diagram for illustrating operations of the vehicle ECU, the vehicle user terminal, the information sharing user terminal, and the bicycle parking lot Bluetooth unit of the wireless communication system according to the first embodiment in normal times and in an emergency;
(14) FIG. **14** is a flowchart for illustrating an operation regarding an emergency notification of the vehicle ECU of the wireless communication system according to the first embodiment;
(15) FIG. **15** is a flowchart for illustrating an operation at a time of an emergency notification of the information sharing user terminal of the wireless communication system according to the first embodiment;
(16) FIG. **16** is a flowchart for illustrating an operation at a time of an emergency notification of the server of the wireless communication system according to the first embodiment;
(17) FIG. **17** is a diagram showing how the wireless communication system according to the first

embodiment tracks a stolen vehicle;
(18) FIG. **18** is a diagram showing an aspect of using a wireless communication system according to a second embodiment;
(19) FIG. **19** is a block diagram showing a schematic configuration of a part of a first vehicle and a schematic configuration of a first vehicle ECU of the wireless communication system according to the second embodiment;
(20) FIG. **20** is a block diagram showing a schematic configuration of a part of a second vehicle and a schematic configuration of a second vehicle ECU of the wireless communication system according to the second embodiment;
(21) FIG. **21** is a block diagram showing a schematic configuration of an information sharing user terminal of the wireless communication system according to the second embodiment;
(22) FIG. **22** is a diagram showing an example of server notification information transmitted to a server from a telecom wireless transmission and reception unit of the information sharing user terminal of the wireless communication system according to the second embodiment;
(23) FIG. **23** is a block diagram showing a schematic configuration of the server of the wireless communication system according to the second embodiment;
(24) FIG. **24** is a diagram showing an example of user notification information transmitted to a vehicle user terminal from a network communication unit of the server of the wireless communication system according to the second embodiment;
(25) FIG. **25** is a sequence diagram for illustrating operations of the first vehicle ECU, the second vehicle ECU, the vehicle user terminal, the information sharing user terminal, and the server of the wireless communication system according to the second embodiment in normal times and in an emergency;
(26) FIG. **26** is a flowchart for illustrating an operation regarding an emergency notification of the second vehicle ECU of the wireless communication system according to the second embodiment;
(27) FIG. **27** is a flowchart for illustrating the operation of the information sharing user terminal of the wireless communication system according to the second embodiment;
(28) FIG. **28** is a diagram showing how the first vehicle of the wireless communication system according to the second embodiment is tracked when the first vehicle is stolen;
(29) FIG. **29** is a diagram showing an overview of operations of a stolen vehicle, a relay vehicle, and information sharing user terminal in a wireless communication system according to a third embodiment;
(30) FIG. **30** is a diagram showing an aspect of using the wireless communication system according to the third embodiment;
(31) FIG. **31** is a block diagram showing a schematic configuration of a part of a second vehicle and a schematic configuration of a second vehicle ECU in the wireless communication system according to the third embodiment;
(32) FIG. **32** is a block diagram showing a schematic configuration of a part of a first vehicle and a schematic configuration of a first vehicle ECU in the wireless communication system according to the third embodiment;
(33) FIG. **33** is a block diagram showing a schematic configuration of the information sharing user terminal of the wireless communication system according to the third embodiment;
(34) FIG. **34** is a diagram showing an example of server notification information transmitted to a server from a telecom wireless transmission and reception unit of the information sharing user terminal of the wireless communication system according to the third embodiment;
(35) FIG. **35** is a block diagram showing a schematic configuration of the server of the wireless communication system according to the third embodiment;
(36) FIG. **36** is a diagram showing an example of user notification information transmitted to a vehicle user terminal from a network communication unit of the server of the wireless communication system according to the third embodiment;

(37) FIG. **37** is a sequence diagram for illustrating operations of the second vehicle ECU, the vehicle user terminal, the information sharing user terminal, and the server when an abnormality occurs in the wireless communication system according to the third embodiment;
(38) FIG. **38** is a sequence diagram for illustrating operations of the second vehicle ECU, the vehicle user terminal, the first vehicle ECU, the information sharing user terminal, and the server when an abnormality occurs in the wireless communication system according to the third embodiment;
(39) FIG. **39** is a flowchart for illustrating an operation regarding an emergency notification of the first vehicle ECU of the wireless communication system according to the third embodiment;
(40) FIG. **40** is a flowchart for illustrating an operation regarding an emergency notification of the information sharing user terminal of the wireless communication system according to the third embodiment;
(41) FIG. **41** is a flowchart for illustrating an operation regarding an emergency notification of the server of the wireless communication system according to the third embodiment;
(42) FIG. **42** is a diagram showing an example of a notification image to a user of the second vehicle of the wireless communication system according to the third embodiment;
(43) FIG. **43** is a block diagram showing a schematic configuration of a part of a vehicle and a schematic configuration of a vehicle ECU of a wireless communication system according to a fourth embodiment;
(44) FIG. **44** is a block diagram showing a schematic configuration of a smartphone of the wireless communication system according to the fourth embodiment;
(45) FIG. **45** is a block diagram showing a schematic configuration of a vehicle electronic key of the wireless communication system according to the fourth embodiment;
(46) FIG. **46** is a diagram showing an aspect of using the wireless communication system according to the fourth embodiment;
(47) FIG. **47** is a sequence diagram showing an operation mainly of the vehicle ECU of the wireless communication system according to the fourth embodiment;
(48) FIG. **48** is a sequence diagram showing an operation mainly of the vehicle ECU of the wireless communication system according to the fourth embodiment;
(49) FIG. **49** is a sequence diagram showing an operation mainly of the vehicle ECU of the wireless communication system according to the fourth embodiment;
(50) FIG. **50** is a sequence diagram showing an operation when a server is provided to improve security in the wireless communication system according to the fourth embodiment;
(51) FIG. **51** is a sequence diagram showing an operation when the server is provided to improve the security in the wireless communication system according to the fourth embodiment;
(52) FIG. **52** is a sequence diagram showing an operation when the server is provided to improve the security in the wireless communication system according to the fourth embodiment;
(53) FIG. **53** is a flowchart for illustrating an operation of the vehicle ECU of the wireless communication system according to the fourth embodiment;
(54) FIG. **54** is a flowchart for illustrating an operation of the smartphone of the wireless communication system according to the fourth embodiment;
(55) FIG. **55** is a flowchart for illustrating an operation of the vehicle ECU when the security is improved in the wireless communication system according to the fourth embodiment;
(56) FIG. **56** is a flowchart for illustrating an operation of the smartphone when the security is improved in the wireless communication system according to the fourth embodiment;
(57) FIG. **57** is a flowchart for illustrating an operation of the server when the security is improved in the wireless communication system according to the fourth embodiment;
(58) FIG. **58** is a diagram showing an aspect of using a wireless communication system according to a fifth embodiment;
(59) FIG. **59** is a diagram showing a case where a smartphone of a vehicle user is present outside a

BLE communication available area of a vehicle ECU in the wireless communication system according to the fifth embodiment;

(60) FIG. **60** is a diagram showing a case where the smartphone of the vehicle user is present in the BLE communication available area of the vehicle ECU in the wireless communication system according to the fifth embodiment;

(61) FIG. **61** is a block diagram showing a schematic configuration of a part of a vehicle and a schematic configuration of the vehicle ECU of the wireless communication system according to the fifth embodiment;

(62) FIG. **62** is a block diagram showing a schematic configuration of the smartphone of the wireless communication system according to the fifth embodiment;

(63) FIG. **63** is a sequence diagram showing an operation mainly of the vehicle ECU in the wireless communication system according to the fifth embodiment;

(64) FIG. **64** is a sequence diagram showing an operation mainly of the vehicle ECU in the wireless communication system according to the fifth embodiment;

(65) FIG. **65** is a flowchart for illustrating an operation of the vehicle ECU in the wireless communication system according to the fifth embodiment; and

(66) FIG. **66** is a flowchart for illustrating an operation of the smartphone of the wireless communication system according to the fifth embodiment.

DESCRIPTION OF EMBODIMENTS

(67) Hereinafter, embodiments specifically disclosing a wireless communication system according to the present disclosure will be described in detail with reference to the drawings as appropriate. However, unnecessarily detailed descriptions may be omitted. For example, detailed descriptions of well-known matters and redundant descriptions of substantially the same configuration may be omitted. This is to avoid the following description from being unnecessarily redundant and facilitate understanding of those skilled in the art. The accompanying drawings and the following description are provided for those skilled in the art to fully understand the present disclosure, and are not intended to limit the subject matters described in the claims.

(68) Hereinafter, preferred embodiments for carrying out the present disclosure will be described in detail with reference to the drawings.

First Embodiment

(69) Hereinafter, a wireless communication system according to a first embodiment will be described with reference to FIGS. **1** to **17**. FIG. **1** is a diagram showing an aspect of using a wireless communication system **1** according to the first embodiment. As shown in FIG. **1**, the wireless communication system **1** according to the first embodiment includes a vehicle electronic control unit (ECU, wireless communication device) **10** mounted on a vehicle **50**, a vehicle user terminal (wireless communication terminal) **11** owned by a user **51** of the vehicle **50**, an information sharing user terminal (wireless communication terminal) **12** owned by a user **52** who shares information with the user **51** of the vehicle **50**, a bicycle parking lot Bluetooth (registered trademark) unit (fixed base station) **13** provided in a bicycle parking lot and connected by wire to a cloud **53**, a server **14** connected by wire to the cloud **53**, and a telecom base station (mobile base station) **15** connected by wire to the cloud **53**. The information sharing user terminal **12** and the bicycle parking lot Bluetooth unit **13** are respectively notification devices. Identification information is given to the vehicle ECU **10**. The identification information can also be considered as identification information of the vehicle **50**. The identification information is information uniquely determined by the vehicle ECU, and may be, for example, a MAC address.

(70) The vehicle ECU **10**, the information sharing user terminal **12**, and the bicycle parking lot Bluetooth unit **13** each have a Bluetooth communication function, and Bluetooth communication is performed between the vehicle ECU **10** and the information sharing user terminal **12** and between the vehicle ECU **10** and the bicycle parking lot Bluetooth unit **13**. The information sharing user terminal **12** has a cellular communication function or a WiFi (registered trademark) communication

function in addition to the Bluetooth communication function. The vehicle user terminal **11** also has a cellular communication function or the WiFi communication function in addition to the Bluetooth communication function.
(71) A smartphone is suitable for the vehicle user terminal **11** and the information sharing user terminal **12** of the wireless communication system **1** according to the first embodiment. Although the wireless communication system **1** according to the first embodiment is applied to the two-wheeled vehicle (motorcycle) **50**, the wireless communication system **1** can of course also be applied to a four-wheeled vehicle.
(72) Next, respective configurations of the vehicle ECU **10**, the vehicle user terminal **11**, the information sharing user terminal **12**, the bicycle parking lot Bluetooth unit **13**, and the server **14** of the wireless communication system **1** according to the first embodiment will be described.
(73) FIG. **2** is a block diagram showing a schematic configuration of a part of the vehicle **50** and a schematic configuration of the vehicle ECU **10**. In FIG. **2**, the vehicle ECU **10** includes a switch detection unit **101**, an IG authentication determination unit **102**, a BT wireless transmission and reception unit **103**, a GPS reception unit **104**, an unauthorized use detection unit (unauthorized use detection circuit) **105**, a vehicle speed detection unit **106**, a clock **107**, a control unit **108**, and antennas **110** and **111**. The switch detection unit **101** detects at least ON/OFF of a glove box switch **501** and ON/OFF of an IG KEY switch **502**. The glove box switch **501** is a switch that detects opening and closing of a glove box (not shown) mounted on the vehicle **50**, and is turned on when the glove box (not shown) is opened and turned off when the glove box is closed. The IG KEY switch **502** is a switch for starting/stopping an engine of the vehicle **50** and turning on/off an electrical system.
(74) The IG authentication determination unit **102** performs IG authentication determination with respect to a key (not shown) inserted into the IG KEY switch **502**. An IG authentication determination result of the IG authentication determination unit **102** is taken into the control unit **108**. The BT wireless transmission and reception unit **103** performs wireless communication conforming to a Bluetooth standard. The antenna **110** for Bluetooth communication is connected to the BT wireless transmission and reception unit **103**. The BT wireless transmission and reception unit **103** is controlled by the control unit **108**. The BT wireless transmission and reception unit **103** and the antenna **110** correspond to a first antenna. The vehicle ECU **10** has one antenna **110**, but may have a plurality thereof.
(75) The GPS reception unit **104** receives a positioning signal transmitted from a global positioning system (GPS) satellite and outputs a second latitude and longitude. The antenna **111** for GPS reception is connected to the GPS reception unit **104**. The second latitude and longitude output from the GPS reception unit **104** is taken into the control unit **108**.
(76) The unauthorized use detection unit **105** includes an acceleration sensor **1051** and detects an abnormality including a theft of the vehicle **50**. That is, when the acceleration sensor **1051** detects an abnormal vibration in the vehicle **50**, the unauthorized use detection unit **105** detects an abnormality including a theft. The unauthorized use detection unit **105** repeatedly performs the detection. A detection result of the unauthorized use detection unit **105** is taken into the control unit **108**. The vehicle speed detection unit **106** detects a vehicle speed from vehicle speed information managed by a vehicle speed management ECU **503** mounted on the vehicle **50**. A detection result of the vehicle speed detection unit **106** is taken into the control unit **108**. The clock **107** outputs a second time indicating a current date and time. The second time output from the clock **107** is taken into the control unit **108**.
(77) The control unit **108** controls each unit of the device, and includes a central processing unit (CPU) (not shown), a read only memory (ROM) storing a program for operating the CPU, and a random access memory (RAM) used for an operation of the CPU. The IG authentication determination unit **102**, the BT wireless transmission and reception unit **103**, the GPS reception unit **104**, the unauthorized use detection unit **105**, the vehicle speed detection unit **106**, and the

clock **107** operate under the control of the control unit **108**. If an abnormality including a theft occurs in the vehicle **50**, the control unit **108** issues an emergency notification to the vehicle user terminal **11** of the vehicle user **51**.
(78) If the unauthorized use detection unit **105** detects unauthorized use (that is, if an abnormality including a theft occurs in the vehicle **50**), the BT wireless transmission and reception unit **103** transmits an advertising packet of Bluetooth including identification information of the vehicle **50**, emergency information, the second time, and the second latitude and longitude from the antenna **110**. In this case, the second time is a time at which the unauthorized use detection unit **105** last detects the unauthorized use. The advertising packet of Bluetooth including the identification information of the vehicle **50**, the emergency information, the second time, and the second latitude and longitude is received by the information sharing user terminal **12** and the bicycle parking lot Bluetooth unit **13**. The advertising packet of Bluetooth including the identification information of the vehicle **50**, the emergency information, the second time, and the second latitude and longitude is called a first advertising packet or emergency advertising.
(79) The first advertising packet transmitted from the vehicle ECU **10** includes at least the identification information of the vehicle **50** and the emergency information. That is, it is unnecessary to include the second time and the second latitude and longitude, and it is sufficient to include the identification information of the vehicle **50** and the emergency information. It is desirable that the first advertising packet is not transmitted only once, but is transmitted twice or more.
(80) If the unauthorized use detection unit **105** does not detect the unauthorized use, the BT wireless transmission and reception unit **103** does not transmit the first advertising packet including the emergency information, but transmits an advertising packet that does not include the emergency information. The advertising packet that does not include the emergency information is called a second advertising packet or normal advertising. The second advertising packet may not be transmitted only once, but may be transmitted twice or more.
(81) When the number of times of transmission of each of the first advertising packet and the second advertising packet is twice or more, it is desirable to make a transmission interval (first transmission interval) of the first advertising packet shorter than a transmission interval (second transmission interval) of the second advertising packet.
(82) When a voltage of a battery (not shown) mounted on the vehicle **50** becomes lower than a predetermined value, it is desirable to make the first transmission interval of the first advertising packet shorter than the second transmission interval of the second advertising packet. That is, the first transmission interval of the first advertising packet is made shorter than that before the voltage of the battery drops. Notification omission can be reduced by shortening the first transmission interval of the first advertising packet that notifies an emergency. If the battery voltage becomes lower than the predetermined value, the transmission of the second advertising packet may be stopped, and at the same time, the second advertising packet may be taken out. The matter that the transmission of the second advertising packet is stopped also means increasing the transmission interval to infinite. The battery may be a vehicle drive battery mounted on the vehicle **50** or a button battery within the vehicle ECU **10**.
(83) If the battery voltage becomes lower than the predetermined value, a countermeasure may be performed in such an order that at first, only the second advertising packet is taken out, and if the battery voltage further drops thereafter, the first advertising packet is taken out, and if the battery voltage drops further thereafter, the transmission of the second advertising packet is stopped.
(84) In the transmission of the first advertising packet, it is desirable to make an electric field strength of a radio wave that transmits the packet greater than an electric field strength of a radio wave that transmits the second advertising packet.
(85) Accordingly, by increasing the number of times of the transmission of the first advertising packet, shortening the transmission interval, or increasing the electric field strength of the radio

wave, a notification at a time of an emergency can be made more reliable. Power saving can be achieved by taking out the second advertising packet or stopping the transmission.

(86) FIGS. **3** and **4** are diagrams showing output waveforms of the first advertising packet and the second advertising packet. In FIG. **3**, when communication between the vehicle ECU **10** and the vehicle user terminal **11** is cut off while the communication is being established by Bluetooth, the vehicle ECU **10** starts the transmission of the second advertising packet (normal advertising). The normal advertising is transmitted at a predetermined interval, and enters into a connection waiting state. If an abnormality such as a theft occurs in the vehicle **50** in this state, the transmission of the first advertising packet (emergency advertising) is started from that time. The transmission interval of the emergency advertising is shorter than the transmission interval of the normal advertising, and the transmission is performed within the transmission interval of the normal advertising.

(87) Next, in FIG. **4**, when a decrease in the battery voltage of the battery (not shown) of the vehicle **50** is detected while the abnormality occurring, the normal advertising is taken out. Then, when the battery voltage further drops, the transmission of the normal advertising is stopped, and the emergency advertising is taken out.

(88) FIG. **5** is a block diagram showing a schematic configuration of the information sharing user terminal **12**. In FIG. **5**, the information sharing user terminal **12** acts as an intermediary to send an emergency notification transmitted from the vehicle ECU **10** to the vehicle user terminal **11** if an abnormality including a theft occurs in the vehicle **50**. The information sharing user terminal **12** includes a GPS reception unit (first location information detection circuit) **121**, an antenna **122**, a clock (first clock) **123**, a BT wireless transmission and reception unit **124**, a telecom transmission and reception unit **125**, antennas **126** and **129**, an advertising determination unit **127**, and a control unit **128**.

(89) The GPS reception unit **121** receives a positioning signal transmitted from the GPS satellite and outputs a first latitude and longitude. The antenna **122** for GPS reception is connected to the GPS reception unit **121**. The clock **123** outputs a first time indicating a current date and time. The BT wireless transmission and reception unit **124** performs wireless communication conforming to a Bluetooth standard. The antenna **126** for Bluetooth communication is connected to the BT wireless transmission and reception unit **124**.

(90) The BT wireless transmission and reception unit **124** receives an advertising packet of Bluetooth via the antenna **126**. The BT wireless transmission and reception unit **124** outputs the received advertising packet to the control unit **128**. The telecom transmission and reception unit **125** performs cellular wireless communication. The antenna **129** for cellular communication is connected to the telecom transmission and reception unit **125**. The advertising determination unit **127** determines whether the advertising packet of Bluetooth received by the BT wireless transmission and reception unit **124** is a normal advertising or an emergency advertising, and outputs the result to control unit **128**.

(91) The control unit **128** controls each unit of the device, and includes a CPU (not shown), a ROM storing a program for operating the CPU, and a RAM used in an operation of the CPU. The ROM also stores an application for sharing vehicle information. The GPS reception unit **121**, the clock **123**, the BT wireless transmission and reception unit **124**, the telecom transmission and reception unit **125**, and the advertising determination unit **127** operate under the control of the control unit **128**. The BT wireless transmission and reception unit **124** and the antenna **126** correspond to a second antenna. The telecom transmission and reception unit **125** and the antenna **129** correspond to a third antenna. The information sharing user terminal **12** includes one antenna **126** and one antenna **129**, but may include a plurality thereof.

(92) The control unit **128** adds the first latitude and longitude and the first time to the advertising packet received by the BT wireless transmission and reception unit **124**, and transmits the obtained packet from the telecom transmission and reception unit **125** to the telecom base station **15** via the antenna **129**. If the advertising packet of Bluetooth received by the BT wireless transmission and

reception unit **124** is the first advertising packet, the packet includes the emergency information, and if the advertising packet of Bluetooth is the second advertising packet, no emergency information is included. If the advertising packet received by the BT wireless transmission and reception unit **124** includes the second latitude and longitude and the second time, the information is also included and transmitted to the telecom base station **15**.

(93) FIG. **6** is a block diagram showing a schematic configuration of the bicycle parking lot Bluetooth unit **13**. In FIG. **6**, the bicycle parking lot Bluetooth unit **13** includes a BT wireless transmission and reception unit **131**, an antenna **132**, a network communication unit **133**, an advertising determination unit **134**, and a control unit **135**. The BT wireless transmission and reception unit **131** performs wireless communication conforming to a Bluetooth standard. The antenna **132** for Bluetooth communication is connected to the BT wireless transmission and reception unit **131**.

(94) The network communication unit **133** connects to the cloud **53** and communicates with the server **14**. The advertising determination unit **134** determines whether the advertising packet of Bluetooth received by the BT wireless transmission and reception unit **131** is a normal advertising or an emergency advertising, and outputs the result. In this case, if the advertising packet of Bluetooth received by the BT wireless transmission and reception unit **131** is the first advertising packet, the advertising packet of Bluetooth is an emergency advertising, and if the advertising packet of Bluetooth is the second advertising packet, the advertising packet of Bluetooth is a normal advertising.

(95) The control unit **135** controls each unit of the device, and includes a CPU (not shown), a ROM storing a program for operating the CPU, and a RAM used in an operation of the CPU. The BT wireless transmission and reception unit **131**, the network communication unit **133**, and the advertising determination unit **134** operate under the control of the control unit **135**. From the determination result of the advertising determination unit **134**, if the advertising packet received by the BT wireless transmission and reception unit **131** is the first advertising packet, the control unit **135** transmits identification information of the vehicle **50** and emergency information from the network communication unit **133** to the server **14** via the cloud **53**, and if the advertising packet is the second advertising packet, the control unit **135** transmits the identification information of the vehicle **50** from the network communication unit **133** to the server **14** via the cloud **53**. If the advertising packet includes the second latitude and longitude and the second time, the information is also included and transmitted to the server **14**. The bicycle parking lot Bluetooth unit **13** has one antenna **132**, but may have a plurality thereof.

(96) FIG. **7** is a block diagram showing a schematic configuration of the server **14**. In FIG. **7**, the server **14** includes a network communication unit **141**, a notification information determination unit **142**, and a control unit **143**. The network communication unit **141** connects to the cloud **53** and communicates with the vehicle user terminal **11**. The notification information determination unit **142** determines whether the notification is a normal notification or an emergency notification based on information received by the network communication unit **141**. If the information received by the network communication unit **141** is the identification information of the vehicle **50** and the emergency information, the emergency is determined, and if no emergency information of the vehicle **50** is present, the notification is determined as the normal notification.

(97) The control unit **143** controls each unit of the device, and includes a CPU (not shown), a ROM storing a program for operating the CPU, a RAM used in an operation of the CPU, and a large-capacity storage device (not shown) such as a hard disk for storing data. The network communication unit **141** and the notification information determination unit **142** operate under the control of the control unit **143**.

(98) When the information of the vehicle **50** is received by the network communication unit **141**, the control unit **143** outputs the information to the notification information determination unit **142** and stores the information in the storage device (not shown). Here, if the information received by

the network communication unit **141** is from the information sharing user terminal **12**, the first and second latitudes and longitudes and the first and second times are stored in addition to the identification information and the emergency information. If the information received by the network communication unit **141** is from the bicycle parking lot Bluetooth unit **13**, the second latitude and longitude and the second time are stored in addition to the identification information and the emergency information. In response to acquiring the information output from the control unit **143**, the notification information determination unit **142** determines whether the notification is a normal notification or an emergency notification based on the information, and outputs the result to the control unit **143**. In the case of the identification information of the vehicle **50** and the emergency information, the control unit **143** determines the result as an emergency, and issues the emergency notification to the vehicle user terminal **11** of the emergency, and in the case of the identification information of the vehicle **50**, the control unit **143** issues the normal notification to the vehicle user terminal **11**. Although WiFi communication is used for notification from the server **14** to the vehicle user terminal **11**, cellular communication may also be used.
(99) Next, the advertising packet of Bluetooth will be described.
(100) Emergency advertising transmission information uses an AD Type: 0xFF Manufacture Specific parameter of advertising data defined by Bluetooth SIG.
(101) FIG. **8** is a diagram showing a packet structure of LE Uncoded PHYs. In FIG. **8**, the AD Type and AD Data of PDU Payload in the packet structure are used for the emergency advertising transmission information. That is, the AD Type: 0xFF Manufacture Specific parameter of the advertising data defined by Bluetooth SIG is used.
(102) FIG. **9** is a diagram showing a data structure of an AD Type: 0xFF. In the AD Type: 0xFF shown in FIG. **9**, a 2-byte Company Identifier Code is defined, and data that follows can be defined independently by a vendor. In the wireless communication system **1** according to the present embodiment, emergency advertising information that includes an “emergency type” and an “occurrence date and time” is defined. Regarding the “emergency type”, for example, definitions include 0x00: no abnormality, and include that 0x01: once impact on vehicle is detected, 0x02: a plurality of times of impacts on vehicle are detected, 0x03: tilt of vehicle is detected, 0x04: glove box is opened, 0x05: acceleration detected in vehicle (may be transporting), 0x06: IG=ON without authentication, 0x07: starter is started without authentication, and 0x08: wheel speed is detected without authentication.
(103) When the unauthorized use detection unit **105** of the vehicle ECU **10** detects that an abnormality occurs in the vehicle **50**, the unauthorized use detection unit **105** transmits emergency advertising information in which the emergency type and the occurrence date and time are set according to a level of the abnormality. The advertising information is also combined with a vendor-specific UUID predefined for an emergency notification device, and the information sharing user terminal **12** and the bicycle parking lot Bluetooth unit **13**, which are notification devices, determine whether the emergency notification device is a device (vehicle ECU **10**) corresponding to the emergency notification based on the UUID.
(104) The information sharing user terminal **12** or the bicycle parking lot Bluetooth unit **13** which receives the emergency advertising information gives a reception date and time, location information, and an ID (for example, a Bluetooth MAC address) for identifying the vehicle **50** to the emergency advertising information, transmits the obtained information to the server **14**.
(105) FIG. **10** is a diagram showing an example of server notification information. As shown in FIG. **10**, the server notification information includes a “type” and a “content”. The “type” includes a “notification device ID”, a “reception date and time”, a “vehicle ID”, “location information”, an “emergency type”, and an “occurrence date and time”. The “notification device ID” includes an ID of the information sharing user terminal **12** and an ID of the bicycle parking lot Bluetooth unit **13**. The “reception date and time” is a date and time at which the notification device (information sharing user terminal **12**, bicycle parking lot Bluetooth unit **13**) receives the emergency advertising

information. The "vehicle ID" is a Bluetooth MAC address used when the vehicle **50** (vehicle ECU **10**) issues a notification. The "location information" is GPS location information possessed by the notification device. The "emergency type" and the "occurrence date and time" are transmitted from the vehicle ECU **10**.

(106) The server **14** confirms that the "notification device ID" and the "vehicle ID" of the received server notification information are IDs registered in advance, and issues the user notification.

(107) FIG. **11** is a diagram showing an example of user notification information. As shown in FIG. **11**, the user notification information includes a "type" and a "content". The "type" includes a "reception date and time", "location information", an "emergency type", and an "occurrence date and time". The "reception date and time" is a date and time at which the notification device (information sharing user terminal **12**, bicycle parking lot Bluetooth unit **13**) receives the emergency advertising information. The "location information" is GPS location information possessed by the notification device. The "emergency type" and the "occurrence date and time" are transmitted from the vehicle ECU **10**.

(108) FIG. **12** is a diagram showing an overview of an operation of the vehicle ECU **10**. In FIG. **12**, the vehicle ECU **10** transitions between an unauthenticated state and an authenticated state according to ON/OFF of the IG KEY switch **502**. That is, when the IG KEY switch **502** is turned from ON to OFF, the vehicle ECU **10** transitions to the unauthenticated state (security ON), and when the IG KEY switch **502** is turned from OFF to ON, the vehicle ECU **10** transitions to the authenticated state (security OFF). When the vehicle ECU **10** is in the unauthenticated state (security ON) and cannot connect to the vehicle user terminal **11** via Bluetooth, the vehicle ECU **10** enters into an abnormality monitoring state. That is, when the vehicle user terminal **11** moves out of a Bluetooth communication range of the vehicle ECU **10**, the connection between the vehicle ECU **10** and the vehicle user terminal **11** by Bluetooth communication is cut off. The vehicle ECU **10** transmits the normal advertising in the abnormality monitoring state.

(109) If the vehicle ECU **10** detects occurrence of an abnormality in the abnormality monitoring state, the vehicle ECU **10** issues an emergency notification. The emergency advertising is transmitted during abnormality detection. After issuing the emergency notification, if the vehicle ECU **10** detects occurrence of an abnormality again, the vehicle ECU **10** issues an emergency notification and transmits the emergency advertising. In this case, if a content of the abnormality is different from the previous time, the emergency type is updated. For example, if the previous time is "once impact to the vehicle" and the current time is "glove box is opened", the emergency type changes, and thus, the emergency type updates.

(110) FIG. **13** is a sequence diagram for illustrating operations of the vehicle ECU **10**, the vehicle user terminal **11**, the information sharing user terminal **12**, and the bicycle parking lot Bluetooth unit **13** in normal times and in an emergency. In FIG. **13**, when the vehicle user terminal **11** is within the Bluetooth communication range of the vehicle ECU **10**, the vehicle user terminal **11** and the vehicle ECU **10** are connected via Bluetooth. In this state, when the vehicle user **51** moves away from the vehicle **50** and the vehicle user terminal **11** moves out of the Bluetooth communication range of the vehicle ECU **10**, the Bluetooth connection between the vehicle user terminal **11** and the vehicle ECU **10** is cut off.

(111) When the Bluetooth connection with the vehicle user terminal **11** is cut off, the vehicle ECU **10** starts the transmission of the second advertising packet (normal advertising). During the transmission of the second advertising packet, the information sharing user terminal **12** and the bicycle parking lot Bluetooth unit **13**, which are notification devices, ignore the information. In this state, if an abnormality occurs in the vehicle **50**, the vehicle ECU **10** starts the transmission of the first advertising packet (emergency advertising). When the information sharing user terminal **12** and the bicycle parking lot Bluetooth unit **13** receive the first advertising packet, the information sharing user terminal **12** and the bicycle parking lot Bluetooth unit **13** respond to the reception from the vehicle ECU **10** and further issue a server notification to the server **14**. The server **14** that

receives the server notification executes a user notification and notifies the vehicle user terminal **11** that an abnormality occurs in the vehicle **50**.

(112) FIG. **14** is a flowchart for illustrating an operation regarding the emergency notification of the vehicle ECU **10** of vehicle **50**. The operation of the vehicle **50** is the operation of the vehicle ECU **10**, so that the subject is not the vehicle **50** but the vehicle ECU **10**. In FIG. **14**, the vehicle ECU **10** first determines whether the IG KEY switch **502** is turned on (step S**10**), if the vehicle ECU **10** determines that the IG KEY switch **502** is turned on (if “YES” in step S**10**), the vehicle ECU **10** transitions to a security OFF state (step S**11**). That is, since the user **51** of the vehicle **50** starts using the vehicle **50**, the security is released. The security OFF state continues until the IG KEY switch **502** is turned off. If the vehicle ECU **10** determines that the IG KEY switch **502** is not turned on (if “NO” in step S**10**), the vehicle ECU **10** transitions to a security ON state (step S**12**). That is, since the user **51** of the vehicle **50** stops using the vehicle **50**, the security is started.

(113) After transitioning to the security ON state, the vehicle ECU **10** first determines whether the Bluetooth connection is established with the vehicle user terminal **11** (step S**13**), if the vehicle ECU **10** determines that the Bluetooth connection is established (if “YES” in step S**13**), the vehicle ECU **10** transitions to a Bluetooth connection state (step S**14**). After the vehicle ECU **10** transitions to the Bluetooth connection state with the vehicle user terminal **11**, the vehicle ECU **10** determines whether occurrence of an abnormality is detected (step S**15**). If the vehicle ECU **10** determines that occurrence of an abnormality is not detected (if “NO” in step S**15**), the vehicle ECU **10** determines whether the IG KEY switch **502** is turned on (step S**16**). If the vehicle ECU **10** determines that the IG KEY switch **502** is not turned on (if “NO” in step S**16**), the vehicle ECU **10** returns to the process in step S**13**, and determines whether the Bluetooth connection is established with the vehicle user terminal **11**. On the other hand, if the vehicle ECU **10** determines that the IG KEY switch **502** is turned on (if “YES” in step S**16**), the vehicle ECU **10** returns to the process in step S**11**, and transitions to the security OFF state.

(114) If the vehicle ECU **10** determines in step S**15** that occurrence of an abnormality is detected (if “YES” in step S**15**), the vehicle ECU **10** notifies a terminal in connection (for example, the vehicle user terminal **11**) of the abnormality detection (step S**17**). After notifying the terminal in connection of the abnormality detection, the vehicle ECU **10** determines whether notification completion is received (step S**18**), and if the vehicle ECU **10** determines that the notification completion is not received (if “NO” in step S**18**), the vehicle ECU **10** returns to the process in step S**17** and notifies the abnormality detection. On the other hand, if the vehicle ECU **10** determines that the notification completion is received (if “YES” in step S**18**), the vehicle ECU **10** returns to the process in step S**16**, and determines whether the IG KEY switch **502** is turned on.

(115) If the vehicle ECU **10** determines in step S**13** that the Bluetooth connection is not established with the vehicle user terminal **11** (if “NO” in step S**13**), the vehicle ECU **10** transitions to a Bluetooth cut-off state (step S**19**). After transitioning to the Bluetooth cut-off state with the vehicle user terminal **11**, the vehicle ECU **10** determines whether occurrence of an abnormality is detected (step S**20**).

(116) If the vehicle ECU **10** determines that occurrence of an abnormality is not detected (if “NO” in step S**20**), the vehicle ECU **10** transmits the normal advertising (second advertising packet) (step S**21**). In this case, the transmission interval is lengthened and a transmission output is decreased. The vehicle ECU **10** starts the transmission of the normal advertising and then stops the transmission of the emergency advertising (first advertising packet) (step S**22**). Here, when the determination in step S**20** is made for the first time, even if the process in step S**22** is performed, it is meaningless because no emergency advertising is transmitted before, but when the determination in step S**20** is made for the second time or more, the emergency advertising may be transmitted before, and thus, if the determination in step S**20** is “NO”, after starting the transmission of the normal advertising, the transmission of emergency advertising is stopped.

(117) After performing the process in step S**22**, the vehicle ECU **10** returns to step S**16** and

determines whether the IG KEY switch **502** is turned on. If the vehicle ECU **10** determines in step S**20** that occurrence of an abnormality is detected (if "YES" in step S**20**), the vehicle ECU **10** transmits the emergency advertising (step S**23**). In this case, the transmission interval is shortened and the transmission output is increased.
(118) After starting the transmission of the emergency advertising, the vehicle ECU **10** determines whether the notification completion is received (step S**24**). If the vehicle ECU **10** determines that the notification completion is received (if "YES" in step S**24**), the vehicle ECU **10** returns to the process in step S**16**, and determines whether the IG KEY switch **502** is turned on. On the other hand, if the vehicle ECU **10** determines that the notification completion is not received (if "NO" in step S**24**), the vehicle ECU **10** determines whether occurrence of a new abnormality is detected (step S**25**). If the vehicle ECU **10** determines that occurrence of a new abnormality is not detected (if "NO" in step S**25**), the vehicle ECU **10** returns to the process in step S**23**, and transmits the emergency advertising. On the other hand, if the vehicle ECU **10** determines that occurrence of a new abnormality is detected (if "YES" in step S**25**), the vehicle ECU **10** updates the emergency type (step S**26**). Then, the vehicle ECU **10** returns to the process in step S**23**, and transmits an emergency advertising with the updated emergency type.
(119) FIG. **15** is a flowchart for illustrating an operation at a time of an emergency notification of the information sharing user terminal **12**. The operation of the information sharing user terminal **12** is an operation of the control unit **128**, but the subject is not the control unit **128** but the information sharing user terminal **12**. In FIG. **15**, the information sharing user terminal **12** first determines whether the advertising packet is received (step S**30**), and if the information sharing user terminal **12** determines that the advertising packet is not received (if "NO" in step S**30**), the information sharing user terminal **12** repeats the determination until the advertising packet is received. If the information sharing user terminal **12** determines that the advertising packet is received (if "YES" in step S**30**), the information sharing user terminal **12** determines whether the received advertising packet is the first advertising packet (emergency advertising) (step S**31**). If the information sharing user terminal **12** determines that the received advertising packet is not the first advertising packet (if "NO" in step S**31**), the information sharing user terminal **12** repeats the determination until the first advertising packet is received.
(120) On the other hand, if the information sharing user terminal **12** determines that the received advertising packet is the first advertising packet (if "YES" in step S**31**), the information sharing user terminal **12** determines whether an emergency advertising type is set (step S**32**). If the information sharing user terminal **12** determines that the emergency advertising type is not set (if "NO" in step S**32**), the information sharing user terminal **12** repeats the determination until the emergency advertising type is set. On the other hand, if the information sharing user terminal **12** determines that the emergency advertising type is set (if "YES" in step S**32**), the information sharing user terminal **12** notifies the server **14** of the emergency advertising (step S**33**).
(121) FIG. **16** is a flowchart for illustrating an operation of the server **14** at the time of the emergency notification. The operation of the server **14** is an operation of the control unit **143**, but the subject is not the control unit **143** but the server **14**. In FIG. **16**, the server **14** first determines whether the server notification is received (step S**40**), and if the server **14** determines that the server notification is not received (if "NO" in step S**40**), the server **14** repeats the determination until the server notification is received. If the server **14** determines that the server notification is received (if "YES" in step S**40**), the server **14** determines whether the notification device ID is registered (step S**41**).
(122) If the server **14** determines that the notification device ID is not registered (if "NO" in step S**41**), the server **14** returns to the process in step S**40**. On the other hand, if the server **14** determines that the notification device ID is registered (if "YES" in step S**41**), the server **14** determines whether the vehicle ID is registered (step S**42**). If the server **14** determines that the vehicle ID is not registered (if "NO" in step S**42**), the server **14** returns to the process in step S**40**. On the other hand,

if the server **14** determines that the vehicle ID is registered (if "YES" in step S**42**), the server **14** performs the user notification (step S**43**) and returns to the process in step S**40**.
(123) FIG. **17** is a diagram showing how the vehicle **50** is tracked when the vehicle **50** is stolen. In FIG. **17**, a reference numeral **298** indicates a communication range of the normal advertising transmitted from the vehicle ECU **10** of the vehicle **50**, and a reference numeral **299** indicates a communication range of the emergency advertising transmitted from the vehicle ECU **10** of the vehicle **50**. If an abnormality occurs in the vehicle **50**, the emergency advertising (first advertising packet) is transmitted. When a theft occurs, the bicycle parking lot Bluetooth unit **13** is within the communication range **299** of the emergency advertising, and thus, the first advertising packet is received by the bicycle parking lot Bluetooth unit **13** and transmitted to the server **14** via a wired line.
(124) Since an installation location of the bicycle parking lot Bluetooth unit **13** is known, a location where the theft occurs can be obtained. When transportation of the vehicle **50** is started by a truck or the like and the information sharing user terminal **12** is present in the middle of transportation, the first advertising packet is received by the information sharing user terminal **12** and transmitted to the server **14** via the cellular communication or the WiFi communication. Thereafter, in the same manner, each time the first advertising packet is received by the information sharing user terminal **12**, the first advertising packet is transmitted to the server **14** via the cellular communication or the WiFi communication. The first advertising packet received by the server **14** is transmitted to the vehicle user terminal **11** via the cellular communication or the WiFi communication.
(125) As described above, in the wireless communication system **1** according to the first embodiment, if the unauthorized use detection unit **105** provided in the vehicle **50** detects that an abnormality occurs in the vehicle **50**, the first advertising packet to which emergency information is added is transmitted from the vehicle **50** to the information sharing user terminal **12**, and the vehicle user terminal **11** is notified of an abnormality in the vehicle **50** from the information sharing user terminal **12** via the server **14**, and thus, the abnormality in the vehicle **50** can be notified to the vehicle user terminal **11** without using a telematics service in the vehicle **50**. Since there is no need to have a communication line or a GPS function to use the telematics service, costs can be reduced. That is, it is possible to notify an abnormality including a theft in the vehicle **50** at a low cost and with low power consumption, and is suitable for a two-wheeled vehicle such as a scooter.
(126) In the wireless communication system **1** according to the first embodiment, the advertising packet of Bluetooth is used to transmit the identification information of the vehicle **50**, the emergency information, and the like, but other types of packets may be used.
(127) Any communication other than Bluetooth may be used to transmit the identification information of the vehicle **50**, emergency information, and the like.
Second Embodiment
(128) Next, a wireless communication system according to a second embodiment will be described.
(129) A wireless communication system **2** according to the second embodiment transmits a first advertising packet to which emergency information is added from a vehicle when an abnormality occurs in the vehicle, and then notifies another vehicle (equipped with a vehicle ECU that can detect an abnormality) in a surrounding area, and notifies a vehicle user terminal of the abnormality and a current location of the vehicle from an information sharing user terminal (smartphone) paired with another vehicle ECU via a server. In a method of directly notifying the information sharing user terminal from the vehicle ECU, if no information sharing user terminal is present near the vehicle ECU, the abnormality cannot be notified, and the notification can be made at a timing when another information sharing user terminal is in close proximity, and thus, the notification is delayed.
(130) The wireless communication system **2** according to the second embodiment can increase a chance of being able to notify an abnormality in the vehicle, and thus, it is possible to prevent the

delay in notifying the vehicle user as much as possible. By increasing the number of another vehicle ECUs that are the same as the vehicle ECU and can detect an abnormality, an area where a theft can be detected expands, and even if a vehicle is stolen from a bicycle parking lot or the like where no information sharing user terminal is present nearby, it is possible to detect the theft at an early stage. By using the information sharing user terminal, connection to the server does not require the vehicle ECU to have a function to use a telematics service, and thus, it is possible to notify an abnormality in a vehicle such as a theft at a low cost and with low power consumption, and it is suitable for a two-wheeled vehicle such as a scooter.

(131) Hereinafter, the wireless communication system according to the second embodiment will be described below with reference to FIGS. **18** to **28**. FIG. **18** is a diagram showing an aspect of using the wireless communication system **2** according to the second embodiment. In FIG. **18**, the same reference numerals are given to components that are common to the components of the wireless communication system **1** according to the first embodiment described above. In FIG. **18**, the wireless communication system **2** according to the second embodiment includes a first vehicle ECU **10**A mounted on a first vehicle **50**A, a second vehicle ECU (wireless communication device) **10**B mounted on a second vehicle **50**B, a vehicle user terminal (wireless communication terminal, smartphone) **11** owned by the user **51** of the first vehicle **50**A, an information sharing user terminal (wireless communication terminal, smartphone) **12**B owned by the user **52** who shares information with the user **51** of the first vehicle **50**A, a server **14**B connected by wire to the cloud **53**, and the telecom base station (mobile base station) **15** connected by wire to the cloud **53**. Identification information is given to the first vehicle ECU **10**A and the second vehicle ECU **10**B. The identification information can also be regarded as identification information of the first vehicle **50**A and identification information of the second vehicle **50**B. The identification information is information that is uniquely determined by the first vehicle ECU **10**A and the second vehicle ECU **10**B, and may be, for example, a MAC address. In FIG. **18**, the second vehicle **50**B is shown as another vehicle other than the first vehicle **50**A, but it is assumed that there is at least one vehicle equipped with the same second vehicle ECU **10**B as that of the second vehicle **50**B.

(132) The first vehicle ECU **10**A, the second vehicle ECU **10**B, the vehicle user terminal **11**, and the information sharing user terminal **12**B each have a Bluetooth (registered trademark) communication function, and Bluetooth communication is available between the first vehicle ECU **10**A and the information sharing user terminal **12**B, between the first vehicle ECU **10**A and the second vehicle ECU **10**B, between the second vehicle ECU **10**B and the information sharing user terminal **12**B, between the first vehicle ECU **10**A and the vehicle user terminal **11**, and between the second vehicle ECU **10**B and the vehicle user terminal **11**. The vehicle user terminal **11** and the information sharing user terminal **12**B have a cellular communication function or a WiFi (registered trademark) communication function in addition to the Bluetooth communication function. Although the wireless communication system **2** according to the second embodiment is applied to the two-wheeled vehicle (motorcycle) **50**A, the wireless communication system **2** can of course also be applied to a four-wheeled vehicle.

(133) Next, respective configurations of the first and second vehicle ECUs **10**A and **10**B, the vehicle user terminal **11**, the information sharing user terminal **12**B, and the server **14**B of the wireless communication system **2** according to the second embodiment will be described.

(134) FIG. **19** is a block diagram showing a schematic configuration of a part of the first vehicle **50**A and a schematic configuration of the first vehicle ECU **10**A. In FIG. **19**, the same reference numerals are given to components that are common to the components of the vehicle ECU **10** shown in FIG. **2**. The first vehicle ECU **10**A includes the switch detection unit **101**, the IG authentication determination unit **102**, the BT wireless transmission and reception unit **103**, the unauthorized use detection unit (unauthorized use detection circuit) **105**, the vehicle speed detection unit **106**, a control unit **108**A, and the antenna **110**. The BT wireless transmission and reception unit **103** and the antenna **110** correspond to a first antenna.

(135) The control unit **108**A controls each unit of the device, and includes a CPU (not shown), a ROM storing a program for operating the CPU, and a RAM used in an operation of the CPU. The IG authentication determination unit **102**, the BT wireless transmission and reception unit **103**, the unauthorized use detection unit **105**, and the vehicle speed detection unit **106** operate under the control of the control unit **108**A.
(136) When an abnormality including a theft occurs in the first vehicle **50**A, the control unit **108**A performs an emergency notification process on the vehicle user terminal **11** of the vehicle user **51**. That is, if the unauthorized use detection unit **105** detects unauthorized use (that is, if an abnormality including a theft occurs in the first vehicle **50**A), the control unit **108**A transmits an advertising packet of Bluetooth including identification information of the vehicle **50**A and emergency information from the BT wireless transmission and reception unit **103** via the antenna **110**. Various kinds of control related to transmission, such as the number of times of transmission of the advertising packet, a transmission interval, and an electric field strength of a transmitted radio wave, are the same as the control in the vehicle ECU **10** shown in FIG. **2**, and are as described with reference to FIGS. **3** and **4**.
(137) FIG. **20** is a block diagram showing a schematic configuration of a part of the second vehicle **50**B and a schematic configuration of the second vehicle ECU **10**B. In FIG. **19**, the same reference numerals are given to components that are common to the components of the vehicle ECU **10** shown in FIG. **2**. The second vehicle ECU **10**B includes the switch detection unit **101**, the IG authentication determination unit **102**, the BT wireless transmission and reception unit **103**, the GPS reception unit **104**, the unauthorized use detection unit **105**, the vehicle speed detection unit **106**, the clock (second clock) **107**, a control unit **108**B, a memory (memory circuit) **109**, and the antenna **110**. The GPS reception unit **104** acquires the second latitude and longitude. The clock **107** acquires the second time. The memory **109** stores at least identification information of the first vehicle **50**A.
(138) The control unit **108**B controls each unit of the device, and includes a CPU (not shown), a ROM storing a program for operating the CPU, and a RAM used in an operation of the CPU. The IG authentication determination unit **102**, the BT wireless transmission and reception unit **103**, the GPS reception unit **104**, the unauthorized use detection unit **105**, the vehicle speed detection unit **106**, the clock **107**, and the memory **109** operate under the control of the control unit **108**B.
(139) If the unauthorized use detection unit **105** detects unauthorized use (that is, if an abnormality including a theft occurs in the second vehicle **50**B), the control unit **108**B transmits an advertising packet of Bluetooth including identification information of the second vehicle **50**B and emergency information from the BT wireless transmission and reception unit **103** via the antenna **110**. The advertising packet of Bluetooth including the emergency information is the first advertising packet. The transmission control on the advertising packet performed by the control unit **108**B is the same as the transmission control performed by the vehicle ECU **10** shown in FIG. **2**, and is as described with reference to FIGS. **3** and **4**.
(140) If the BT wireless transmission and reception unit **103** receives an advertising packet of Bluetooth including the identification information of the first vehicle **50**A and the emergency information via the antenna **110**, the control unit **108**B transmits a predetermined packet of Bluetooth including the identification information of the first vehicle **50**A, the emergency information, the second time, and the second latitude and longitude via the antenna **110**. The predetermined packet only needs to include at least the identification information of the first vehicle **50**A and the emergency information. The predetermined packet is a packet that is transmitted without specifying a communication partner. The predetermined packet is defined as a third advertising packet (normal advertising, corresponding to the second advertising packet). The third advertising packet is transmitted twice or more.
(141) FIG. **21** is a block diagram showing a schematic configuration of the information sharing user terminal **12**B. In FIG. **21**, the same reference numerals are given to components that are

common to the components of the information sharing user terminal **12** shown in FIG. **5**. The information sharing user terminal **12**B includes the GPS reception unit (first location information detection circuit) **121**, the BT wireless transmission and reception unit **124**, the telecom transmission and reception unit **125**, the advertising determination unit **127**, the clock (first clock) **123**, and the antennas **122**, **126**, and **129**. The GPS reception unit **121** acquires the first latitude and longitude. The clock **123** acquires the first time. The BT wireless transmission and reception unit **124** is capable of wireless communication with the second vehicle **50**B conforming to the Bluetooth standard. The antenna **129** can be used for cellular communication. The BT wireless transmission and reception unit **124** and the antenna **126** correspond to the third antenna. The telecom transmission and reception unit **125** and the antenna **129** correspond to a fourth antenna.
(142) A control unit **128**B controls each unit of the device, and includes a CPU (not shown), a ROM storing a program for operating the CPU, and a RAM used in an operation of the CPU. The GPS reception unit **121**, the clock **123**, the BT wireless transmission and reception unit **124**, the telecom transmission and reception unit **125**, and the advertising determination unit **127** operate under the control of the control unit **128**B.
(143) If the BT wireless transmission and reception unit **124** receives the predetermined packet of Bluetooth including the identification information of the first vehicle **50**A, the emergency information, the second time, and the second latitude and longitude via the antenna **126**, the control unit **128**B transmits the identification information of the first vehicle **50**A, the emergency information, the second time, and the second latitude and longitude from the telecom transmission and reception unit **125** via the antenna **129**. The transmission only needs to include at least the identification information of the first vehicle **50**A and the emergency information.
(144) FIG. **22** is a diagram showing server notification information transmitted to the server **14**B from the telecom transmission and reception unit **125** of the information sharing user terminal **12**B. As shown in FIG. **22**, the server notification information generated by the telecom transmission and reception unit **125** is obtained by adding a “reliability of location information” to the server notification information generated by the telecom transmission and reception unit **125** of the wireless communication system **1** according to the first embodiment described above. The reliability of the location information is the number of times and a time of control (engine running) after the data is first acquired. The “engine running” is control for starting the engine. The “time” is a time required from the start of the engine to the notification of the abnormality detection to the terminal in connection.
(145) FIG. **23** is a block diagram showing a schematic configuration of the server **14**B. In FIG. **23**, the same reference numerals are given to components that are common to the components of the server **14** shown in FIG. **7**. The server **14**B includes the network communication unit **141**, the notification information determination unit **142**, and a control unit **143**B. FIG. **24** is a diagram showing user notification information transmitted to the vehicle user terminal **11** from the network communication unit **141** of the server **14**B. As shown in FIG. **24**, the user notification information is obtained by adding the “reliability of location information” to the user notification information of the wireless communication system **1** according to the first embodiment described above. The reliability of the location information is the number of times and a time of control (control for starting the engine, engine running) after the data is first acquired.
(146) FIG. **25** is a sequence diagram for illustrating operations of the first vehicle ECU **10**A, the second vehicle ECU **10**B, the vehicle user terminal **11**, the information sharing user terminal **12**B, and the server **14**B in normal times and in an emergency. In FIG. **25**, when the vehicle user terminal **11** is within a Bluetooth communication range of the first vehicle ECU **10**A, the vehicle user terminal **11** and the first vehicle ECU **10**A are connected via Bluetooth. In this state, when the vehicle user **51** moves away from the first vehicle **50**A and the vehicle user terminal **11** moves out of the Bluetooth communication range of the first vehicle ECU **10**A, the Bluetooth connection between the vehicle user terminal **11** and the first vehicle ECU **10**A is cut off.

(147) When the Bluetooth connection with the vehicle user terminal **11** is cut off, the first vehicle ECU **10**A starts the transmission of the third advertising packet. The second vehicle ECU **10**B ignores the information while the third advertising packet is being transmitted. In this state, if an abnormality occurs in the first vehicle **50**A, the first vehicle ECU **10**A starts the transmission of the first advertising packet (emergency advertising).
(148) If the second vehicle ECU **10**B receives the first advertising packet, the second vehicle ECU **10**B responds to the reception from the first vehicle ECU **10**A, and further establishes the Bluetooth connection with the information sharing user terminal **12**B. The information sharing user terminal **12**B establishing the Bluetooth connection with the second vehicle ECU **10**B issues a server notification to the server **14**B. The server **14**B that receives the server notification executes a user notification and notifies the vehicle user terminal **11** that an abnormality occurs in the first vehicle **50**A.
(149) Next, operations regarding the emergency notification of the first vehicle ECU **10**A and the second vehicle ECU **10**B will be described. However, the operation regarding the emergency notification of the first vehicle ECU **10**A is similar to the operation regarding the emergency notification of the vehicle ECU **10** of the wireless communication system **1** according to the first embodiment described above, and thus, a description is omitted here, and only the operation of the second vehicle ECU **10**B is described.
(150) FIG. **26** is a flowchart for illustrating the operation regarding the emergency notification of the second vehicle ECU **10**B. The present process includes a process of adding the reliability of the location information to each of the server notification information shown in FIG. **22** and the user notification information shown in FIG. **24**. The control unit **108**B is provided with a counter (not shown) for counting the number of times of starting the engine and a counter (not shown) for counting a time after the engine is started. The operation of the second vehicle **50**B is an operation of the control unit **108**B, but a subject is not the control unit **108**B but the second vehicle ECU **10**B.
(151) In FIG. **26**, the second vehicle ECU **10**B first determines whether the emergency advertising is received (step S**50**), and if the second vehicle ECU **10**B determines that the emergency advertising is not received (if “NO” in step S**50**), the second vehicle ECU **10**B repeats the present process until the second vehicle ECU **10**B determines that the emergency advertising is received. If the second vehicle ECU **10**B determines that the emergency advertising is received (if “YES” in step S**50**), the second vehicle ECU **10**B retains the emergency information (step S**51**).
(152) After retaining the emergency information, the second vehicle ECU **10**B determines whether an engine of the vehicle **50**B is being started (step S**52**). If the second vehicle ECU **10**B determines that the engine of the vehicle **50**B is being started (if “YES” in step S**52**), the second vehicle ECU **10**B determines whether the Bluetooth connection is established with the notification device, that is, the information sharing user terminal **12**B (step S**53**). If the second vehicle ECU **10**B determines that the Bluetooth connection is not established with the information sharing user terminal **12**B (if “NO” in step S**53**), the second vehicle ECU **10**B returns to step S**52**, and if the second vehicle ECU **10**B determines that the Bluetooth connection is established with the information sharing user terminal **12**B (if “YES” in step S**53**), the second vehicle ECU **10**B notifies the terminal in Bluetooth connection, that is, the information sharing user terminal **12**B of the abnormality detection (step S**54**). In this case, the number of times of starting the engine is “0” and the time after the engine is started is “0”. That is, since the engine is being started, the number of times of starting the engine is “0”. After notifying the information sharing user terminal **12**B of the abnormality detection, the second vehicle ECU **10**B returns to the process in step S**50**.
(153) If the second vehicle ECU **10**B determines in step S**52** that the engine is not started (if “NO” in step S**52**), the second vehicle ECU **10**B determines whether the Bluetooth connection is established with the notification device, that is, the information sharing user terminal **12**B (step S**55**). If the second vehicle ECU **10**B determines that the Bluetooth connection is established with

the information sharing user terminal **12**B (if "YES" in step S**55**), the second vehicle ECU **10**B determines whether the engine start control on the vehicle **50**B is performed (step S**56**), and if the second vehicle ECU **10**B determines that the engine start control is not performed (if "NO" in step S**56**), the second vehicle ECU **10**B repeats the present process until the engine start control is performed.

(154) If the second vehicle ECU **10**B determines that the engine start control is performed (if "YES" in step S**56**), the second vehicle ECU **10**B adds "1" to the number of times of starting the engine (step S**57**). That is, when the engine is started while the engine is in a stop state, the number of times of starting the engine becomes a value in which "1" is added. Next, the second vehicle ECU **10**B notifies the terminal in Bluetooth connection, that is, the information sharing user terminal **12**B of the abnormality detection (step S**58**). In this case, the number of times of starting the engine is "n" and the time after the engine is started is "0". After notifying the information sharing user terminal **12**B of the abnormality detection, the second vehicle ECU **10**B returns to the process in step S**50**.

(155) If the second vehicle ECU **10**B determines in step S**55** described above that the Bluetooth connection is not established with the notification device, that is, the information sharing user terminal **12**B (if "NO" in step S**55**), the second vehicle ECU **10**B determines whether the engine start control is performed (step S**59**), and if the second vehicle ECU **10**B determines that the engine start control is not performed (if "NO" in step S**59**), the second vehicle ECU **10**B returns to step S**55**. If the second vehicle ECU **10**B determines in step S**59** that the engine start control is performed (if "YES" in step S**59**), "1" is added to the number of times of starting the engine, and an engine start time counter is turned "ON" (step S**60**).

(156) After turning the engine start time counter "ON" in step S**60**, the second vehicle ECU **10**B determines whether the Bluetooth connection is established with the notification device, that is, the information sharing user terminal **12**B (step S**61**), and if the second vehicle ECU **10**B determines that the Bluetooth connection is established with the information sharing user terminal **12**B (if "YES" in step S**61**), the second vehicle ECU **10**B notifies the terminal in Bluetooth connection, that is, the information sharing user terminal **12**B of the abnormality detection (step S**64**). In this case, the number of times of starting the engine is "n" and the time after the engine is started is "n". After notifying the information sharing user terminal **12**B of the abnormality detection, the second vehicle ECU **10**B returns to the process in step S**50**.

(157) If the second vehicle ECU **10**B determines in step S**61** described above that the Bluetooth connection is not established with the notification device, that is, the information sharing user terminal **12**B (if "NO" in step S**61**), the second vehicle ECU **10**B determines whether the engine is turned off (step S**62**). If the second vehicle ECU **10**B determines that the engine is not turned off (if "NO" in step S**62**), the second vehicle ECU **10**B returns to step S**61**, and if the second vehicle ECU **10**B determines that the engine is turned off (if "YES" in step S**62**), the second vehicle ECU **10**B clears the engine start time counter (step S**63**), and returns to step S**55**.

(158) Next, the operation of the information sharing user terminal **12**B will be described. The operation of the server **14**B is the same as the operation of the server **14** of the wireless communication system **1** according to the first embodiment described above, and thus, descriptions thereof will be omitted.

(159) FIG. **27** is a flowchart for illustrating the operation of the information sharing user terminal **12**B. The operation of the information sharing user terminal **12**B is an operation of the control unit **128**B, but the subject is not the control unit **128**B but the information sharing user terminal **12**B. In FIG. **27**, the information sharing user terminal **12**B first determines whether the Bluetooth connection is established with the second vehicle ECU **10**B (step S**70**), and if the information sharing user terminal **12**B determines that the Bluetooth connection is not established with the second vehicle ECU **10**B (if "NO" in step S**70**), the information sharing user terminal **12**B repeats the present process until the Bluetooth connection is established with the second vehicle ECU **10**B.

(160) If the information sharing user terminal **12**B determines that the Bluetooth connection is established with the second vehicle ECU **10**B (if “YES” in step S**70**), the information sharing user terminal **12**B determines whether the second vehicle ECU **10**B retains the emergency advertising information (step S**71**). If the information sharing user terminal **12**B determines that the second vehicle ECU **10**B does not retain the emergency advertising information (if “NO” in step S**71**), the information sharing user terminal **12**B returns to step S**70**, and if the information sharing user terminal **12**B determines that the second vehicle ECU **10**B retains the emergency advertising information (if “YES” in step S**71**), the information sharing user terminal **12**B notifies the server **14**B of the emergency advertising information (step S**72**). After notifying the server **14**B of the emergency advertising information, the process returns to step S**70**.
(161) FIG. **28** is a diagram showing how the first vehicle **50**A is tracked when the first vehicle **50**A is stolen. In FIG. **28**, the reference numeral **298** indicates a communication range of the normal advertising transmitted from the first vehicle ECU **10**A of the first vehicle **50**A, and the reference numeral **299** indicates a communication range of the emergency advertising transmitted from the first vehicle ECU **10**A of the first vehicle **50**A. If an abnormality occurs in the first vehicle **50**A, the emergency advertising (first advertising packet) is transmitted from the vehicle ECU **10**A. If another vehicles such as a second vehicle **50**B-**1** and a third vehicle **50**B-**2** are present within the communication range of the emergency advertising when a theft occurs, the emergency advertising is received by these vehicles **50**B-**1** and **50**B-**2** and transmitted by Bluetooth. If the emergency advertising transmitted from the second vehicle **50**B-**1** and the third vehicle **50**B-**2** is received by information sharing user terminals **12**B-**1** and **12**B-**2**, identification information of the first vehicle **50**A, emergency information, and the like included in the emergency advertising are transmitted from these terminals **12**B-**1** and **12**B-**2** via the cellular communication. Since a current location of each of the second vehicle **50**B-**1** and the third vehicle **50**B-**2** is obtained, the location where the theft occurs can be acquired.
(162) If the transportation of the first vehicle **50**A is started by a truck or the like, and a new vehicle (referred to as a fourth vehicle **50**B-**3**) is present in the middle of the transportation, the emergency advertising is received by the fourth vehicle **50**B-**3** and transmitted by Bluetooth. If the emergency advertising transmitted from the fourth vehicle **50**B-**3** is received by a new information sharing user terminal (referred to as an information sharing user terminal **12**B-**3**), identification information of the first vehicle **50**A, emergency information, and the like included in the emergency advertising are transmitted from the new information sharing user terminal via the cellular communication. Since the current location is acquired by the fourth vehicle **50**B-**3**, the location in the middle of transportation can be acquired.
(163) Thereafter, in the same way, if a new vehicle (referred to as a fifth vehicle **50**B-**4**) is present in the middle of the transportation of the first vehicle **50**A, the fifth vehicle **50**B-**4** receives the emergency advertising, and then transmits the emergency advertising via Bluetooth. If the emergency advertising transmitted from the fifth vehicle **50**B-**4** is received by a new information sharing user terminal (referred to as an information sharing user terminal **12**B-**4**), identification information of the first vehicle **50**A, emergency information, and the like included in the emergency advertising are transmitted from the new information sharing user terminal via the cellular communication. Since the current location is acquired by the fifth vehicle **50**B-**4**, the location in the middle of transportation can be acquired.
(164) The identification information of the first vehicle **50**A, the emergency information, and the like transmitted by the cellular communication from the information sharing user terminal **12**B-**1** to the information sharing user terminal **12**B-**4** are transmitted to the server **14**B via a wired line, and are transmitted from the server **14**B to the vehicle user terminal **11** of the vehicle **50**A.
(165) As described above, in the wireless communication system **2** according to the second embodiment, if the unauthorized use detection unit **105** provided in the first vehicle **50**A detects that an abnormality occurs in the first vehicle **50**A, the first advertising packet to which the

emergency information is added is transmitted from the first vehicle **50**A to notify the second vehicle **50**B presenting nearby, and the information sharing user terminal **12**B paired with the second vehicle **50**B notifies the vehicle user terminal **11** of the abnormality in the first vehicle **50**A via the server **14**B, and thus, the vehicle user terminal **11** can be notified of the abnormality in the first vehicle **50**A without using the telematics service. Since there is no need to have a communication line or a GPS function to use the telematics service, an increase in the cost can be kept to a minimum. A chance of being able to notify an abnormality in the first vehicle **50**A is increased, and thus, it is possible to prevent the delay in notifying the vehicle user as much as possible. By increasing the number of another vehicles, other than the second vehicle **50**B, which are the same as the first vehicle ECU **50**A and can detect an abnormality, an area where a theft can be detected expands, and even if a vehicle is stolen from a bicycle parking lot or the like where the information sharing user terminal **12**B is not present nearby, it is possible to detect the theft at an early stage. That is, it is possible to notify an abnormality including a theft in the first vehicle **50**A with low cost and low power consumption, and is suitable for a two-wheeled vehicle such as a scooter.

Third Embodiment

(166) Next, a wireless communication system according to a third embodiment will be described.

(167) A wireless communication system **3** according to the third embodiment is configured such that an advertising packet of Bluetooth (registered trademark) to which emergency information (predetermined information) is added is transmitted from a stolen vehicle (stolen vehicle), at least the emergency information and a distance to the stolen vehicle is transmitted to the server if an information sharing user terminal (smartphone) receives the advertising packet, and when the information sharing user terminal is not present in an area where the advertising packet transmitted from the stolen vehicle can be received, and another vehicle (relay vehicle) that can receive the advertising packet is present, the another vehicle transmits at least the emergency information, a distance (first distance) to the information sharing user terminal paired with the another vehicle, and the distance (second distance) to the stolen vehicle to the server via the information sharing user terminal. In the measurement of the distance, the Bluetooth communication and/or ultra wide band (UWB) ranging technology is used.

(168) FIG. **29** is a diagram showing an overview of operations of a second vehicle (stolen vehicle) **50**E, a first vehicle (relay vehicle) **50**F, and an information sharing user terminal **12**C or **12**D in the wireless communication system **3** according to the third embodiment. In FIG. **29**, when an abnormality occurs in the second vehicle **50**E, the advertising packet of Bluetooth to which the emergency information is added is transmitted. This advertising packet reaches an emergency advertising area EA, and if the information sharing user terminal **12**C is present within the emergency advertising area EA, the information sharing user terminal **12**C receives the advertising packet. By receiving the advertising packet, the information sharing user terminal **12**C acquires a distance to the second vehicle **50**E, and transmits at least the acquired distance and the emergency information from the second vehicle **50**E to the server.

(169) On the other hand, if the information sharing user terminal **12**C is not present in the emergency advertising area EA and the first vehicle **50**F is present therein, the first vehicle **50**F receives the advertising packet from the second vehicle **50**E. By receiving the advertising packet, the first vehicle **50**F acquires a distance to the information sharing user terminal **12**D that is paired with the first vehicle **50**F and is present in a smartphone connection area SA, and the distance to the second vehicle **50**E, and transmits at least the acquired the first and second distances and the emergency information from the second vehicle **50**E to the server via the information sharing user terminal **12**D.

(170) As described above, when using a communication method with a long communication distance such as Bluetooth LE communication, if only the location information is used, an area where the stolen vehicle is assumed to be present becomes wider, and it is assumed to be difficult

to actually find the stolen vehicle. The wireless communication system **3** according to the third embodiment acquires the distance between the stolen vehicle and the information sharing user terminal, the distance between the information sharing user terminal and the relay vehicle, and the distance between the stolen vehicle and the relay vehicle, and thus, a range of the area where the stolen vehicle is assumed to be present can be estimated, and the stolen vehicle can be easily found.
(171) Hereinafter, the wireless communication system according to the third embodiment will be described below with reference to FIGS. **30** to **42**. FIG. **30** is a diagram showing an aspect of using the wireless communication system **3** according to the third embodiment. In FIG. **18**, the same reference numerals are given to components that are common to the components of the wireless communication system **1** according to the first embodiment described above. In FIG. **30**, the wireless communication system **3** according to the third embodiment includes a second vehicle ECU **10**E mounted on the second vehicle **50**E, a first vehicle ECU **10**F mounted on the first vehicle **50**F, the vehicle user terminal (wireless communication terminal, smartphone) **11** owned by the user **51** of the second vehicle **50**E, the information sharing user terminal (wireless communication terminal, smartphone) **12**D owned by the user **52** who shares information with the user **51** of the second vehicle **50**E, a server **14**D connected by wire to the cloud **53**, and the telecom base station (mobile base station) **15** connected by wire to the cloud **53**.
(172) Identification information is provided to the second vehicle ECU **10**E and the first vehicle ECU **10**F. The identification information can also be regarded as identification information of the second vehicle **50**E and identification information of the first vehicle **50**F. The identification information is information that is uniquely determined by the second vehicle ECU **10**E and the first vehicle ECU **10**F, and may be, for example, a MAC address. In FIG. **30**, the first vehicle **50**F is shown as another vehicle other than the second vehicle **50**E, and it is assumed that there is at least one vehicle equipped with the first vehicle ECU **10**F same as that of the first vehicle **50**F.
(173) The second vehicle ECU **10**E, the first vehicle ECU **10**F, the vehicle user terminal **11**, and the information sharing user terminal **12**D each have a Bluetooth communication function, and Bluetooth communication is available between the second vehicle ECU **10**E and the information sharing user terminal **12**D, between the second vehicle ECU **10**E and the first vehicle ECU **10**F, between the first vehicle ECU **10**F and the information sharing user terminal **12**D, between the second vehicle ECU **10**E and the vehicle user terminal **11**, and between the first vehicle ECU **10**F and the vehicle user terminal **11**. The vehicle user terminal **11** and the information sharing user terminal **12**C have a cellular communication function or a WiFi (registered trademark) communication function in addition to the Bluetooth communication function. Although the wireless communication system **3** according to the third embodiment is applied to the two-wheeled vehicle (motorcycle) **50**E, the wireless communication system **3** can of course also be applied to a four-wheeled vehicle.
(174) Next, respective configurations of the first and second vehicle ECUs **10**F and **10**E, the vehicle user terminal **11**, the information sharing user terminal **12**D, and the server **14**D of the wireless communication system **3** according to the third embodiment will be described.
(175) FIG. **31** is a block diagram showing a schematic configuration of a part of the second vehicle **50**E and a schematic configuration of the second vehicle ECU **10**E. In FIG. **31**, the same reference numerals are given to components that are common to the components of the first vehicle ECU **10**A shown in FIG. **19**. The second vehicle ECU **10**E includes the switch detection unit **101**, the IG authentication determination unit **102**, the BT wireless transmission and reception unit (second wireless communication circuit) **103**, the unauthorized use detection unit (unauthorized use detection circuit) **105**, the vehicle speed detection unit **106**, the clock (clock circuit) **107**, a control unit **108**E, and the antenna **110**. The unauthorized use detection unit **105** includes an acceleration sensor **1051**. The clock **107** acquires a time.
(176) The control unit **108**E controls each unit of the device, and includes a CPU (not shown), a ROM storing a program for operating the CPU, and a RAM used in an operation of the CPU. The

IG authentication determination unit **102**, the BT wireless transmission and reception unit **103**, the unauthorized use detection unit **105**, the vehicle speed detection unit **106**, and the clock **107** operate under the control of the control unit **108**E.

(177) When an abnormality including a theft occurs in the second vehicle **50**E, the control unit **108**E performs an emergency notification process on the vehicle user terminal **11** of the vehicle user **51**. That is, if the unauthorized use detection unit **105** detects unauthorized use (that is, if an abnormality including a theft occurs in the second vehicle **50**E), the control unit **108**E transmits an advertising packet of Bluetooth including identification information of the second vehicle **50**E and emergency information (second information) from the BT wireless transmission and reception unit **103** via the antenna **110**. Various kinds of control related to transmission, such as the number of times of transmission of the advertising packet, a transmission interval, and an electric field strength of a transmitted radio wave, are the same as the control in the vehicle ECU **10** shown in FIG. **2**, and are as described with reference to FIGS. **3** and **4**.

(178) FIG. **32** is a block diagram showing a schematic configuration of a part of the first vehicle **50**F and a schematic configuration of the first vehicle ECU **10**F. In FIG. **32**, the same reference numerals are given to components that are common to the components of the second vehicle ECU **10**B shown in FIG. **20**. The first vehicle ECU **10**F includes the switch detection unit **101**, the IG authentication determination unit **102**, the BT wireless transmission and reception unit (first wireless communication circuit) **103**, the unauthorized use detection unit **105**, the vehicle speed detection unit **106**, the clock (clock circuit) **107**, a control unit **108**F, the antenna **110**, and an advertising detection unit **112**. The clock **107** acquires a time. The advertising detection unit **112** detects an advertising packet from a received signal of the BT wireless transmission and reception unit **103**.

(179) The control unit **108**F controls each unit of the device, and includes a CPU (not shown), a ROM storing a program for operating the CPU, and a RAM used in an operation of the CPU. The IG authentication determination unit **102**, the BT wireless transmission and reception unit **103**, the unauthorized use detection unit **105**, the vehicle speed detection unit **106**, the clock **107**, and the advertising detection unit **112** operate under the control of the control unit **108**F.

(180) If the unauthorized use detection unit **105** detects unauthorized use (that is, if an abnormality including a theft occurs in the first vehicle **50**F), the control unit **108**F transmits an advertising packet of Bluetooth including identification information of the first vehicle **50**F and emergency information (predetermined information, first information) from the BT wireless transmission and reception unit **103** via the antenna **110**. The transmission control on the advertising packet performed by the control unit **108**F is the same as the transmission control performed by the vehicle ECU **10** shown in FIG. **2**, and is as described with reference to FIGS. **3** and **4**.

(181) That is, if the BT wireless transmission and reception unit **103** receives the advertising packet of Bluetooth including the identification information of the second vehicle **50**E and the emergency information (second information) via the antenna **110**, the control unit **108**F repeatedly measures an estimated distance to the second vehicle **50**E. A time at which the last advertising packet is received and the estimated distance (second distance) to the second vehicle **50**E are held. Then, if the control unit **108**F detects that the notification device, that is, the information sharing user terminal **12**D enters a communication area, the control unit **108**F establishes the Bluetooth connection with the information sharing user terminal **12**D and measures an estimated distance (first distance) to the information sharing user terminal **12**D. Further, the control unit **108**F calculates an estimated distance between the notification device, that is, the information sharing user terminal **12**D, and the stolen vehicle, that is, the second vehicle **50**E (first distance+second distance). The control unit **108**F notifies the server **14**D of the first distance, the second distance, the first distance+the second distance, and the second information via the information sharing user terminal **12**D.

(182) FIG. **33** is a block diagram showing a schematic configuration of the information sharing user terminal **12**D (**12**C). In FIG. **21**, the same reference numerals are given to components that are

common to the components of the information sharing user terminal **12** shown in FIG. **5**. The information sharing user terminal **12**D (**12**C) includes the GPS reception unit (location information acquisition circuit) **121**, the clock (clock circuit) **123**, the BT wireless transmission and reception unit **124**, the telecom transmission and reception unit **125**, the advertising determination unit **127**, a control unit **128**E, and the antennas **122**,**126**, and **129**. The GPS reception unit **121** acquires location information. The clock **123** acquires a time. The BT wireless transmission and reception unit **124** is capable of wireless communication conforming to the Bluetooth standard with the first vehicle **50**F or the second vehicle **50**E. The antenna **129** can be used for cellular communication.
(183) The control unit **128**E controls each unit of the device, and includes a CPU (not shown), a ROM storing a program for operating the CPU, and a RAM used in an operation of the CPU. The GPS reception unit **121**, the clock **123**, the BT wireless transmission and reception unit **124**, the telecom transmission and reception unit **125**, and the advertising determination unit **127** operate under the control of the control unit **128**E.
(184) If the BT wireless transmission and reception unit **124** receives the first distance, the second distance, the first distance+the second distance, and the second information via the antenna **126**, the control unit **128**E adds the time acquired from the clock **123** and the location information acquired from the GPS reception unit **121**, and transmits the obtained information from the telecom transmission and reception unit **125** to the server **14**D via the antenna **129**. In response to the advertising packet of Bluetooth transmitted from the second vehicle **50**E, the control unit **128**E measures the estimated distance to the stolen vehicle, that is, the second vehicle **50**E, adds the measurement result, the time acquired from the clock **123** when the advertising packet is received, and the location information acquired from the GPS reception unit **121**, and transmits the obtained information from the telecom transmission and reception unit **125** to the server **14**D via the antenna **129**.
(185) FIG. **34** is a diagram showing an example of server notification information transmitted to the server **14**D from the telecom transmission and reception unit **125** of the information sharing user terminal **12**C. As shown in FIG. **34**, the server notification information generated in the telecom transmission and reception unit **125** includes a “type” and a “content”. The “type” includes a “notification device ID”, a “reception date and time”, a “vehicle ID”, “location information”, “estimated distance information”, an “emergency type”, and an “occurrence date and time”. The “notification device ID” is an ID of the information sharing user terminal **12**D. The “reception date and time” is a date and time at which the notification device (information sharing user terminal **12**D) receives emergency advertising information. The “vehicle ID” is a Bluetooth MAC address used when the second vehicle **50**E (second vehicle ECU **10**E) issues a notification. The “location information” is GPS location information possessed by the notification device. The “estimated distance information” is information on an estimated distance between the notification device and the stolen vehicle. The stolen vehicle is the second vehicle **50**E. The “emergency type” is an emergency type notified by the second vehicle **50**E. The “occurrence date and time” is an occurrence date and time notified by the second vehicle **50**E. The “emergency type” and the “occurrence date and time” are transmitted from the second vehicle ECU **10**E of the second vehicle **50**E.
(186) The server **14**D confirms that the “notification device ID” and the “vehicle ID” of the received server notification information are IDs registered in advance, and issues the user notification.
(187) FIG. **35** is a block diagram showing a schematic configuration of the server **14**D. In FIG. **23**, the same reference numerals are given to components that are common to the components of the server **14** shown in FIG. **7**. The server **14**D includes the network communication unit **141**, the notification information determination unit **142**, and a control unit **143**C. FIG. **36** is a diagram showing an example of user notification information transmitted to the vehicle user terminal **11** from the network communication unit **141** of the server **14**D. As shown in FIG. **11**, the user

notification information includes a “type” and a “content”. The “type” includes a “reception date and time”, “location information”, “estimated distance information”, an “emergency type”, and an “occurrence date and time”.

(188) The “reception date and time” is a date and time at which the notification device receives emergency advertising information. The “location information” is GPS location information possessed by the notification device. The “estimated distance information” is information on an estimated distance between the notification device and the stolen vehicle. The stolen vehicle is the second vehicle **50**E. The “emergency type” is an emergency type notified by the second vehicle **50**E. The “occurrence date and time” is an occurrence date and time notified by the second vehicle **50**E. The “emergency type” and the “occurrence date and time” are transmitted from the second vehicle ECU **10**E of the second vehicle **50**E.

(189) An operation of the second vehicle ECU **10**E is as described with reference to FIG. **12** in the first embodiment. The first vehicle ECU **10**F, which relays information from the second vehicle **50**E to the information sharing user terminal **12**D, can also operate in the same manner as the second vehicle ECU **10**E. That is, when the first vehicle ECU **10**F is stolen, the operation shown in FIG. **12** is performed.

(190) FIG. **37** is a sequence diagram for illustrating operations of the second vehicle ECU **10**E, the vehicle user terminal **11**, the information sharing user terminal **12**C, and the server **14**D when an abnormality occurs in the wireless communication system **3** according to the third embodiment. In FIG. **37**, when an abnormality occurs in the second vehicle **50**E (when the second vehicle **50**E is stolen), the second vehicle ECU **10**E starts the transmission of an emergency advertising (advertising packet of Bluetooth). In this case, even if the second vehicle ECU **10**E transmits an emergency advertising, the emergency advertising is not received unless the information sharing user terminal **12**C is present within a Bluetooth communication range of the second vehicle ECU **10**E.

(191) When the information sharing user terminal **12**C enters the Bluetooth communication range of the second vehicle ECU **10**E during movement of the stolen second vehicle **50**E or due to the information sharing user terminal **12**C in close proximity to the second vehicle **50**E, the emergency advertising transmitted from the second vehicle ECU **10**E is received by the information sharing user terminal **12**C. In response to receiving the emergency advertising, the information sharing user terminal **12**C measures an estimated distance between the notification device (information sharing user terminal **12**C) and the second vehicle **50**E which is a stolen vehicle. After measuring the estimated distance between the notification device and the second vehicle **50**E, the information sharing user terminal **12**C notifies the server **14**D of information including the estimated distance to the second vehicle **50**E. The server **14**D notifies the vehicle user terminal **11** of the second vehicle **50**E of the information notified from the information sharing user terminal **12**C. The user **51** of the second vehicle **50**E can confirm the information notified from the server **14**D on the vehicle user terminal **11** and know that his or her vehicle, that is, the second vehicle **50**E, is stolen. The second vehicle **50**E may measure the estimated distance between the information sharing user terminal **12**C and the second vehicle **50**E, and notify the information sharing user terminal **12**C of the result.

(192) FIG. **38** is a sequence diagram for illustrating operations of the second vehicle ECU **10**E, the vehicle user terminal **11**, the first vehicle ECU **10**F, the information sharing user terminal **12**D, and the server **14**D when an abnormality occurs in the wireless communication system **3** according to the third embodiment. In FIG. **38**, when the second vehicle **50**E is stolen, the second vehicle ECU **10**E starts the transmission of the emergency advertising (advertising packet of Bluetooth). In this case, if the first vehicle **50**F enters the Bluetooth communication range of the second vehicle ECU **10**E, the first vehicle ECU **10**F of the first vehicle **50**F receives the emergency advertising from the second vehicle ECU **10**E.

(193) In response to receiving the emergency advertising transmitted from the second vehicle ECU

10E of the second vehicle **50**E, the first vehicle ECU **10**F measures the estimated distance (second distance) between the first vehicle **50**F, which is a reception vehicle, and the second vehicle **50**E, which is the stolen vehicle. Since either the first vehicle **50**F or the second vehicle **50**E may be moving, the first vehicle ECU **10**F repeatedly measures the distance. The first vehicle ECU **10**F temporarily retains the time of measurement each time the distance is measured. Then, when the first vehicle ECU **10**F cannot receive the emergency advertising, the first vehicle ECU **10**F retains the last received time and estimated distance. If either the first vehicle **50**F or the second vehicle **50**E is moving, the first vehicle ECU **10**F may move out of the Bluetooth communication range of the second vehicle ECU **10**E, and thus, this case becomes a time of last reception. The second vehicle ECU **10**E may measure the estimated distance between the first vehicle **50**F and the second vehicle **50**E, which is a stolen vehicle, and notify the first vehicle ECU **10**F of the result.
(194) If the information sharing user terminal **12**D enters a Bluetooth communication range of the first vehicle ECU **10**F, the information sharing user terminal **12**D establishes the Bluetooth connection with the first vehicle ECU **10**F. After establishing the Bluetooth connection with the first vehicle ECU **10**F, the information sharing user terminal **12**D measures the estimated distance (first distance) between the notification device, that is, the information sharing user terminal **12**D, and the reception vehicle, that is, the first vehicle **50**F. Next, the information sharing user terminal **12**D calculates the estimated distance between the notification device, that is, the information sharing user terminal **12**D, and the stolen vehicle, that is, the second vehicle **50**E. That is, the first distance and the second distance are added. This calculation may be performed by the first vehicle ECU **10**F and notified to the information sharing user terminal **12**D.
(195) The information sharing user terminal **12**D notifies the server **14**D of the information including the estimated distance to the stolen vehicle (second vehicle **50**E). The server **14**D notifies the vehicle user terminal **11** of the second vehicle **50**E of the information notified from the information sharing user terminal **12**D. The user **51** of the second vehicle **50**E can confirm the information notified from the server **14**D on the vehicle user terminal **11** and know that his or her vehicle, that is, the second vehicle **50**E, is stolen.
(196) The operation regarding the emergency notification of the second vehicle ECU **10**E of the second vehicle **50**E is the same as the operation regarding the emergency notification of the vehicle ECU **10** of the vehicle **50** according to the first embodiment shown in FIG. **14**, and thus, a description thereof is omitted here. The first vehicle ECU **10**F of the first vehicle **50**F also performs the same operation as the operation regarding the emergency notification of the second vehicle ECU **10**E.
(197) FIG. **39** is a flowchart showing an operation corresponding to the sequence diagram of FIG. **38**. That is, FIG. **39** is a flowchart for illustrating the operation regarding the emergency notification of the first vehicle ECU **10**F. An operation of the first vehicle ECU **10**F is an operation of the control unit **108**F, but the subject is not the control unit **108**F but the first vehicle ECU **10**F.
(198) In FIG. **39**, the first vehicle ECU **10**F first determines whether the emergency advertising is received (step S**220**), and if the first vehicle ECU **10**F determines that the emergency advertising is not received (if “NO” in step S**220**), the first vehicle ECU **10**F repeats the present process until the first vehicle ECU **10**F determines that the emergency advertising is received. If the first vehicle ECU **10**F determines that the emergency advertising is received (if “YES” in step S**220**), the first vehicle ECU **10**F measures the estimated distance to the stolen vehicle, that is, the second vehicle **50**E (step S**221**).
(199) After measuring the estimated distance to the stolen vehicle, the first vehicle ECU **10**F retains the last received time and the estimated distance (step S**222**). Next, the first vehicle ECU **10**F determines whether the connection is established with the notification device, that is, the information sharing user terminal **12**D (step S**223**), and if the first vehicle ECU **10**F determines that the connection is established with the notification device (if “YES” in step S**223**), the first vehicle ECU **10**F measures the estimated distance to the notification device (step S**224**). After

measuring the estimated distance to the notification device, the first vehicle ECU **10**F calculates the estimated distance between the notification device and the stolen vehicle (step S**225**). That is, the first vehicle ECU **10**F adds the estimated distance to the stolen vehicle and the estimated distance to the notification device.

(200) After calculating the estimated distance between the notification device and the stolen vehicle, the first vehicle ECU **10**F notifies the terminal in connection, that is, the information sharing user terminal **12**D of the abnormality detection (step S**226**). This notification includes the estimated distance between the notification device, that is, the information sharing user terminal **12**D and the stolen vehicle (second vehicle **50**E). After performing the process in step S**226**, the first vehicle ECU **10**F returns to the process in step S**220**.

(201) On the other hand, if the first vehicle ECU **10**F determines in the process of step S**223** that the connection is not established with the notification device, that is, the information sharing user terminal **12**D (if "NO" in step S**223**), the first vehicle ECU **10**F determines whether the engine start control on the first vehicle **50**F is performed (step S**227**). If the first vehicle ECU **10**F determines that the engine start control on the first vehicle **50**F is not performed (if "NO" in step S**227**), the first vehicle ECU **10**F returns to step S**223**. If the first vehicle ECU **10**F determines that the engine start control on the first vehicle **50**F is performed (if "YES" in step S**227**), the first vehicle ECU **10**F determines whether the Bluetooth connection is established with the notification device, that is, the information sharing user terminal **12**D (step S**228**).

(202) If the first vehicle ECU **10**F determines that the Bluetooth connection is not established with the information sharing user terminal **12**D (if "NO" in step S**228**), the first vehicle ECU **10**F continues the present process until the first vehicle ECU **10**F determines that the Bluetooth connection is established with the information sharing user terminal **12**D. If the first vehicle ECU **10**F determines that the Bluetooth connection is established with the information sharing user terminal **12**D (if "YES" in step S**228**), the first vehicle ECU **10**F notifies the terminal in connection, that is, the information sharing user terminal **12**D of the abnormality detection (step S**229**). This notification includes the estimated distance between the notification device, that is, the information sharing user terminal **12**D and the stolen vehicle (second vehicle **50**E). In this case, the estimated distance has a large error. After performing the process in step S**229**, the first vehicle ECU **10**F returns to the process in step S**220**.

(203) FIG. **40** is a flowchart for illustrating an operation regarding an emergency notification of the notification device, that is, the information sharing user terminal **12**D. An operation of the information sharing user terminal **12**D is an operation of the control unit **128**E, but the subject is not the control unit **128**E but the information sharing user terminal **12**D. In FIG. **40**, the information sharing user terminal **12**D determines whether the connection is established with the first vehicle ECU **10**F of the first vehicle **50**F (step S**240**), and if the information sharing user terminal **12**D determines that the connection is not established with the first vehicle ECU **10**F (if "NO" in step S**240**), the information sharing user terminal **12**D continues the present process until the information sharing user terminal **12**D determines that the connection is established with the first vehicle ECU **10**F.

(204) If the information sharing user terminal **12**D determines that the connection is established with the first vehicle ECU **10**F (if "YES" in step S**240**), the information sharing user terminal **12**D determines whether the first vehicle ECU **10**F retains the emergency advertising information (step S**241**). If the information sharing user terminal **12**D determines that the first vehicle ECU **10**F does not retain the emergency advertising information (if "NO" in step S**241**), the information sharing user terminal **12**D continues the present process until the information sharing user terminal **12**D determines that the first vehicle ECU **10**F retains the emergency advertising information.

(205) If the information sharing user terminal **12**D determines that the first vehicle ECU **10**F retains the emergency advertising information (if "YES" in step S**241**), the information sharing user terminal **12**D measures the estimated distance to the reception vehicle, that is, the first vehicle

50F (step S**242**). After measuring the estimated distance to the reception vehicle, the information sharing user terminal **12**D calculates the estimated distance to the stolen vehicle, that is, the second vehicle **50**E (step S**243**). After calculating the estimated distance to the stolen vehicle, the information sharing user terminal **12**D notifies the server **14**D of the information including the calculated estimated distance (step S**244**). After notifying the server **14**D of the information including the estimated distance, the information sharing user terminal **12**D returns to the process in step S**240**. Although the first vehicle ECU **10**F and the information sharing user terminal **12**D measure the distance between each other independently, either one may perform measurement and notify the other of the results. The calculation of the distances to the stolen vehicle, that is, the second vehicle **50**E, is performed by the first vehicle ECU **10**F and the information sharing user terminal **12**D, respectively, and it is of course possible to notify the information sharing user terminal **12**D of the result of the distance measurement performed by the first vehicle ECU **10**F.
(206) FIG. **41** is a flowchart for illustrating an operation regarding emergency notification of the server **14**D. An operation of the server **14**D is an operation of the control unit **143**C, but the subject is not the control unit **143**C but the server **14**D. In FIG. **41**, the server **14**D determines whether a server notification from the information sharing user terminal **12**D is received (step S**250**), and if the server **14**D determines that the server notification is not received (if “NO” in step S**250**), the server **14**D continues the present process until the server **14**D determines that the server notification is received. After determining that the server notification is received, the server **14**D determines whether the notification device ID is registered (step S**251**), and if the server **14**D determines that the notification device ID is not registered (if “NO” in step S**251**), the server **14**D returns to the process in step S**250**.
(207) If the server **14**D determines that the notification device ID is registered (if “YES” in step S**251**), the server **14**D determines whether the vehicle ID is registered (step S**252**), and if the server **14**D determines that the vehicle ID is not registered (if “NO” in step S**252**), the server **14**D returns to the process of step S**250**. On the other hand, if the server **14**D determines that the vehicle ID is registered (if “YES” in step S**252**), the server **14**D performs the user notification (step S**253**) and returns to the process in step S**250**.
(208) FIG. **42** is a diagram showing an example of a notification image to the user **51** of the second vehicle **50**E. In FIG. **42**, marks MK-1 to MK-3 indicate locations, and a last time confirmed at each location is displayed directly above each mark MK. When the vehicle (second vehicle **50**E) moves, an error in estimating the vehicle location is large, and thus, the marks MK are displayed, for example, in a blinking manner. A dashed circle CI surrounding the mark MK-3 indicates an error circle based on the estimated distance information.
(209) As described above, in the wireless communication system **3** according to the third embodiment, if the second vehicle **50**E detects an abnormality such as a theft, the second vehicle ECU **10**E of the second vehicle **50**E transmits the advertising packet of Bluetooth to which the emergency information is added, if the information sharing user terminal **12**C receives this advertising packet, the distance to the second vehicle **50**E is measured, the measurement result, and the emergency information, the time at which the emergency information is received, the location at which the emergency information is received, and the like are transmitted to the server **14**D, and on the other hand, if the information sharing user terminal **12**C is not present in the area where the advertising packet transmitted from the second vehicle ECU **10**E of the second vehicle **50**E can be received, and the first vehicle **50**F that can receive the advertising packet is present, the first vehicle ECU **10**F of the first vehicle **50**F relays, and transmits the emergency information, the distance (first distance) to the information sharing user terminal **12**D paired with the first vehicle ECU **10**F, the distance (second distance) to the second vehicle **50**E, which is a stolen vehicle, and the like to the server **14**D via the information sharing user terminal **12**D, and thus, the range of the area where the stolen vehicle is assumed to be present can be estimated, and the stolen vehicle can be easily found.

Fourth Embodiment

(210) Next, a wireless communication system according to a fourth embodiment will be described.

(211) In recent years, each driver of a car or a motorcycle owns a smartphone, which is suitable for identifying individual owners. Since the smartphone can communicate with a cloud server through mobile communication (4th generation mobile communication standard (4G)/long term evolution (LTE)/5th generation mobile communication standard (5G), and the like) it is possible to have a function of a vehicle key, such as periodically authenticating with a password or forcibly disabling use from the cloud server. Even if a smartphone with such a function is stolen, it is possible to construct a system that permanently prevents anyone other than an official owner from using the function of the vehicle key. That is, a theft detection method that prioritizes the smartphone of the official owner as an official owner authentication method is useful.

(212) Hereinafter, a wireless communication system **4** according to the fourth embodiment will be described below with reference to FIGS. **43** to **57**. The wireless communication system **4** according to the fourth embodiment basically includes a vehicle **50**C and a vehicle ECU (wireless communication device) **10**C shown in FIG. **43**, a smartphone (second wireless communication terminal) **11**C shown in FIG. **44**, and an electronic key (first wireless communication terminal) **16** shown in FIG. **45**. Both the smartphone **11**C and the electronic key **16** can be used as keys for the vehicle **50**C equipped with the vehicle ECU **10**C. The vehicle ECU **10**C, the smartphone **11**C, and the electronic key **16** each have a Bluetooth (registered trademark) communication function, and Bluetooth communication is available between the smartphone **11**C and the vehicle ECU **10**C and between the electronic key **16** and the vehicle ECU **10**C. The smartphone **11**C has a cellular communication function or a WiFi (registered trademark) communication function in addition to the Bluetooth communication function. Identification information is given to the vehicle ECU **10**C. The identification information can also be considered as identification information of the vehicle **50**C. The identification information is information uniquely determined by the vehicle ECU, and may be, for example, a MAC address. The electronic key is also called Key FOB (or simply FOB).

(213) The vehicle ECU **10**C has a function of detecting that the vehicle is stolen if the smartphone **11**C is not in close proximity to (that is, away from) the vehicle **50**C for a certain period of time (for example, one week) and the vehicle is operated, and transitioning to a vehicle theft mode. Although the wireless communication system **4** according to the fourth embodiment is applied to the two-wheeled vehicle (motorcycle) **50**C, the wireless communication system **4** can of course also be applied to a four-wheeled vehicle. Hereinafter, configurations of the vehicle ECU **10**C, the smartphone **11**C, and the electronic key **16** of the wireless communication system **4** according to the fourth embodiment will be sequentially described.

(214) FIG. **43** is a block diagram showing a schematic configuration of a part of the vehicle **50**C and a schematic configuration of the vehicle ECU **10**C. In FIG. **43**, the same reference numerals are given to components that are common to the components of the vehicle ECU **10** (see FIG. **2**) of the wireless communication system **1** according to the first embodiment described above. In FIG. **43**, the vehicle ECU **10**C of the wireless communication system **4** according to the fourth embodiment includes a switch detection unit **101**C, the IG authentication determination unit **102**, the BT wireless transmission and reception unit (first wireless communication circuit) **103**, the unauthorized use detection unit **105**, the vehicle speed detection unit **106**, a control unit **108**C, and the antenna **110**. The vehicle ECU **10**C has one antenna **110**, but may have a plurality thereof.

(215) The switch detection unit **101**C detects ON/OFF of the glove box switch **501** and the IG KEY switch **502**, and detects ON/OFF of a main switch (operation unit) **504** provided in the vehicle **50**C. The main switch **504** is a switch for starting an engine of the vehicle **50**C. The BT wireless transmission and reception unit (first wireless communication circuit) **103** performs the wireless communication conforming to the Bluetooth standard (first wireless method) as described above.

(216) The control unit **108**C controls each unit of the device, and includes a CPU (not shown), a ROM storing a program for operating the CPU, and a RAM used in an operation of the CPU. The

IG authentication determination unit **102**, the BT wireless transmission and reception unit **103**, the unauthorized use detection unit **105**, and the vehicle speed detection unit **106** operate under the control of the control unit **108**C. If an abnormality including a theft occurs in the vehicle **50**C, the control unit **108**C issues an emergency notification to the smartphone **11**C of the vehicle user **51**. When communicating with the smartphone **11**C, the control unit **108**C registers the smartphone **11**C. Details of an operation of the control unit **108**C will be described later.
(217) FIG. **44** is a block diagram showing a schematic configuration of the smartphone **11**C. In FIG. **44**, the same reference numerals are given to components that are common to the components of the information sharing user terminal (smartphone) **12** (see FIG. **5**) of the wireless communication system **1** according to the first embodiment described above. As shown in FIG. **44**, the smartphone **11**C includes the GPS reception unit **121**, the BT wireless transmission and reception unit **124**, the telecom transmission and reception unit **125**, the advertising determination unit **127**, an authentication unit **137**, the antennas **122**, **126**, and **129**, and a control unit **128**C. The smartphone **11**C includes one antenna **126** and one antenna **129**, but may include a plurality thereof.
(218) The BT wireless transmission and reception unit (third wireless communication circuit) **124** performs the wireless communication conforming to the Bluetooth standard (first wireless method) as described above. The telecom transmission and reception unit (fourth wireless communication circuit) **125** performs wireless communication conforming to a cellular communication method (second wireless method) different from the Bluetooth standard communication method (first wireless method).
(219) The authentication unit **137** authenticates the user of the smartphone **11**C. The authentication unit **137** has at least one authentication function among password authentication, fingerprint authentication, and face authentication, and outputs an authentication result to the control unit **128**C. The control unit **128**C controls each unit of the device, and includes a CPU (not shown), a ROM storing a program for operating the CPU, and a RAM used in an operation of the CPU. The GPS reception unit **121**, the BT wireless transmission and reception unit **124**, the telecom transmission and reception unit **125**, the advertising determination unit **127**, and the authentication unit **137** operate under the control of the control unit **128**C. When the control unit **128**C communicates with the vehicle ECU **10**C, the control unit **128**C transmits information for registering the control unit **128**C in the vehicle ECU **10**C.
(220) FIG. **45** is a block diagram showing a schematic configuration of the electronic key **16** of the vehicle **50**C. In FIG. **45**, the electronic key **16** includes a BT wireless transmission and reception unit **161**, a key switch **162**, a light emitting diode (LED) **163**, a control unit **164**, and an antenna **165**. The BT wireless transmission and reception unit (second wireless communication circuit) **161** performs the wireless communication conforming to the Bluetooth standard (first wireless method) as described above. The antenna **165** for Bluetooth communication is connected to the BT wireless transmission and reception unit **161**. The BT wireless transmission and reception unit **161** is controlled by the control unit **164**. The key switch **162** is a switch that allows the vehicle **50**C to be operated. A switch signal of the key switch **162** is taken into the control unit **164**.
(221) The LED **163** is an indicator that indicates an operating state of the key switch **162**. The control unit **164** controls each unit of the device, and includes a CPU (not shown), a ROM storing a program for operating the CPU, and a RAM used in an operation of the CPU. The BT wireless transmission and reception unit **161** and the LED **163** operate under the control of the control unit **164**. The control unit **164** receives a switch signal that allows the vehicle **50**C to be operated from the key switch **162**, and transmits a Bluetooth signal that allows the vehicle **50**C to be operated from the BT wireless transmission and reception unit **161**. The control unit **164** controls lighting of the LED **163** by receiving the switch signal from the key switch **162** that allows the vehicle **50**C to be operated.
(222) The electronic key **16** has one antenna **165**, but may have a plurality thereof.

(223) Next, operations of the vehicle ECU **10**C and the smartphone **11**C will be described.
(224) The vehicle ECU **10**C becomes operable when the BT wireless transmission and reception unit (first wireless communication circuit) **103** and the BT wireless transmission and reception unit (second wireless communication circuit) **161** of the electronic key **16** enter into a first communication state. The vehicle ECU **10**C becomes operable when the BT wireless transmission and reception unit (first wireless communication circuit) **103** and the BT wireless transmission and reception unit (third wireless communication circuit) **124** of the smartphone **11**C enter into a second communication state.
(225) After the BT wireless transmission and reception unit (first wireless communication circuit) **103** of the vehicle **50**C and the BT wireless transmission and reception unit (third wireless communication circuit) **124** of the smartphone **11**C enter into the second communication state, if the BT wireless transmission and reception unit (first wireless communication circuit) **103** and the BT wireless transmission and reception unit (third wireless communication circuit) **124** are not in the second communication state for a predetermined time (for example, one week) or longer, and if the main switch (operation unit) **504** of the vehicle **50**C detects a predetermined operation, the vehicle ECU **10**C transmits an emergency advertising (packet including emergency information) from the BT wireless transmission and reception unit (first wireless communication circuit) **103** of the vehicle **50**C. The process of transmitting the emergency advertising is as described in the wireless communication system **1** according to the first embodiment.
(226) The process of transmitting the emergency advertising is performed even when the BT wireless transmission and reception unit (first wireless communication circuit) **103** and the BT wireless transmission and reception unit (second wireless communication circuit) **161** of the electronic key **16** (first wireless communication terminal) are in the first communication state. That is, even when the BT wireless transmission and reception unit (first wireless communication circuit) **103** and the BT wireless transmission and reception unit (second wireless communication circuit) **161** of the electronic key **16** (first wireless communication terminal) are in the first communication state, after the BT wireless transmission and reception unit (first wireless communication circuit) **103** and the BT wireless transmission and reception unit (third wireless communication circuit) **124** of the smartphone (second wireless communication terminal) **11**C enter into the second communication state, if the BT wireless transmission and reception unit (first wireless communication circuit) **103** and the BT wireless transmission and reception unit (third wireless communication circuit) **124** are not in the second communication state for a predetermined time (for example, one week) or longer, and if the main switch (operation unit) **504** of the vehicle **50**C detects a predetermined operation, the vehicle ECU **10**C transmits an emergency advertising from the BT wireless transmission and reception unit (first wireless communication circuit) **103**.
(227) In response to receiving a signal to stop the telecom transmission and reception unit **125** entering into the second communication state, the control unit **128**C of the smartphone **11**C stops at least the BT wireless transmission and reception unit **124** entering into the second communication state. The control unit **128**C operates the authentication unit **137** to perform authentication every time the smartphone **11**C is used, and if the authentication is successful, the control unit **128**C continues to cause at least the BT wireless transmission and reception unit **124** to enter into the second communication state, and if the authentication fails, the control unit **128**C stops at least the BT wireless transmission and reception unit **124** entering into the second communication state. Although the communication state is divided into the first communication state and the second communication state, these states may be the same state.
(228) FIG. **46** is a diagram showing an aspect of using the wireless communication system **4** according to the fourth embodiment. In FIG. **46**, the wireless communication system **4** according to the fourth embodiment includes the vehicle **50**C, the vehicle ECU **10**C, the smartphone **11**C, and the electronic key **16** which are described above. The vehicle **50**C belongs to the vehicle user **51** who is an official owner, and is equipped with the vehicle ECU **10**C. The smartphone **11**C and the

electronic key **16** belong to the vehicle user **51**. In addition to the smartphone **11**C and the electronic key **16** of the vehicle user **51**, it is assumed that a smartphone **11**E and an electronic key **16**E of another person other than the vehicle user **51** are present. The Bluetooth communication available area (BLE communication available area) of the vehicle ECU **10**C is indicated by a reference numeral **300**, and a proximity detection area is indicated by a reference numeral **301**.

(229) FIGS. **47** to **49** are sequence diagrams showing the operation mainly of the vehicle ECU **10**C. In FIG. **47**, the vehicle ECU **10**C periodically performs connection authentication and area detection when the vehicle ECU **10**C is not connected to the smartphone **11**C or the electronic key **16**. Then, if a vehicle operation request is output from the main switch **504** of the vehicle **50**C while the proximity of the smartphone **11**C of the vehicle user **51** is not confirmed for a predetermined time or longer, the vehicle ECU **10**C detects a theft and prohibits a vehicle operation. Furthermore, the transmission of the emergency advertising is started.

(230) In FIG. **48**, the vehicle ECU **10**C detects the proximity of the smartphone **11**E or the electronic key **16**E of another person other than the vehicle user **51** while performing the connection authentication and the area detection, and in this state, if the vehicle operation request is output from the main switch **504** of the vehicle **50**C, the vehicle ECU **10**C continues to prohibit the vehicle operation and continues the transmission of the emergency advertising.

(231) In FIG. **49**, when the vehicle ECU **10**C detects the proximity of the smartphone **11**C of the vehicle user **51** while performing the connection authentication and the area detection, the vehicle ECU **10**C cancels the theft detection and also cancels vehicle operation prohibition. Furthermore, the transmission of the emergency advertising is stopped. Then, when the vehicle operation request is output from the main switch **504** of the vehicle **50**C, the vehicle ECU **10**C turns on power of the vehicle **50**C.

(232) FIGS. **50** to **52** are sequence diagrams showing operations when a server **14**C having an authentication function is provided to achieve high security. The server **14**C has an authentication function in addition to having the same function as the server **14** shown in FIG. **7**. In FIG. **50**, the vehicle ECU **10**C periodically performs the connection authentication and the area detection, and when in a state where the proximity of the smartphone **11**C of the vehicle user **51** is not confirmed for a predetermined time or longer, if the vehicle operation request is output from the main switch **504** of the vehicle **50**C, the vehicle ECU **10**C detects a theft and prohibits the vehicle operation. Furthermore, the transmission of the emergency advertising is started.

(233) In FIG. **51**, the smartphone **11**C of the vehicle user **51** performs server authentication with the server **14**C, and if the server authentication is NG, the server **14**C notifies the smartphone **11**C that the server authentication is NG. In response to receiving the server authentication NG notification, the smartphone **11**C notifies the vehicle ECU **10**C that the server authentication is NG. After receiving the server authentication NG notification, the vehicle ECU **10**C detects the proximity of the smartphone **11**C of the owner whose server authentication is NG, and in this state, when the vehicle operation request is output from the main switch **504** of the vehicle **50**C, the vehicle ECU **10**C continues to prohibit the vehicle operation and further continues the transmission of the emergency advertising.

(234) In FIG. **52**, the smartphone **11**C of the vehicle user **51** performs the server authentication with the server **14**C, and if the server authentication is OK, the server **14**C notifies the smartphone **11**C that the server authentication is OK. When the smartphone **11**C receives the server authentication OK notification and the vehicle ECU **10**C is connected to the smartphone **11**C through the connection authentication and the area detection, the vehicle ECU **10**C notifies the smartphone **11**C that the server authentication is OK. When the vehicle ECU **10**C detects the proximity of the smartphone **11**C of the owner whose server authentication is OK while receiving the server authentication OK notification from the smartphone **11**C, the vehicle ECU **10**C cancels the theft detection and cancels the vehicle operation prohibition. Furthermore, the transmission of the emergency advertising is stopped. Then, when the vehicle operation request is output from the

main switch **504** of the vehicle **50**C, the vehicle ECU **10**C turns on the power of the vehicle **50**C.
(235) Next, the operations of the vehicle ECU **10**C and the smartphone **11**C will be described.
(236) FIG. **53** is a flowchart for illustrating the operation of the vehicle ECU **10**C. In FIG. **53**, the vehicle ECU **10**C first determines whether the smartphone **11**C of the vehicle user **51** is in close proximity for a predetermined time (for example, one week) or longer (step S**80**), and if the vehicle ECU **10**C determines that the smartphone **11**C of the vehicle user **51** is not in close proximity for a predetermined time or longer (that is, if proximity is not confirmed, and if "NO" in step S**80**), the vehicle ECU **10**C determines whether a vehicle operation request is present (step S**81**). If the vehicle ECU **10**C determines that the smartphone **11**C of the vehicle user **51** is in close proximity for a predetermined time or longer (that is, if proximity is confirmed, and if "YES" in step S**80**), the vehicle ECU **10**C repeats the present process until the vehicle ECU **10**C determines that the smartphone **11**C of the vehicle user **51** is not in close proximity for a predetermined time or longer.
(237) If the vehicle ECU **10**C determines in step S**81** that the vehicle operation request is not present (if "NO" in step S**81**), the vehicle ECU **10**C repeats the present process until the vehicle ECU **10**C determines that the vehicle operation request is present. If the vehicle ECU **10**C determines that the vehicle operation request is present (if "YES" in step S**81**), the vehicle ECU **10**C detects a theft (step S**82**), and then prohibits the vehicle operation (step S**83**). Furthermore, the emergency advertising is transmitted (step S**84**).
(238) After transmitting the emergency advertising, the vehicle ECU **10**C determines whether the smartphone **11**C of the vehicle user **51** is in close proximity (step S**85**), and if the vehicle ECU **10**C determines that the smartphone **11**C is not in close proximity (if "NO" in step S**85**), the vehicle ECU **10**C returns to the process in step S**84** and continues to transmit the emergency advertising. If the vehicle ECU **10**C determines that the smartphone **11**C of the vehicle user **51** is in close proximity (if "YES" in step S**85**), the vehicle ECU **10**C cancels the theft detection (step S**86**), then cancels the vehicle operation prohibition (step S**87**), and further stops the transmission of the emergency advertising (step S**88**). After stopping the transmission of the emergency advertising, the vehicle ECU **10**C returns to the process in step S**80**.
(239) FIG. **54** is a flowchart for illustrating the operation of the smartphone **11**C. The operation of the smartphone **11**C is an operation of the control unit **128**C, but the subject is not the control unit **128**C but the smartphone **11**C. In FIG. **54**, the smartphone **11**C is not connected to the vehicle ECU **10**C (step S**90**), and then determines whether the smartphone **11**C is present in the BLE communication available area **300** (step S**91**). If the smartphone **11**C determines that the smartphone **11**C is not present in the BLE communication available area **300** (if "NO" in step S**91**), the smartphone **11**C returns to the process in step S**90**, and if the smartphone **11**C determines that the smartphone **11**C is present in the BLE communication available area **300** (if "YES" in step S**91**), the smartphone **11**C performs the connection authentication with the vehicle ECU **10**C (step S**92**), and establishes connection with the vehicle ECU **10**C (step S**93**). Next, the smartphone **11**C determines whether itself (smartphone **11**C) is present in the BLE communication available area **300** (step S**94**), and if the smartphone **11**C determines that the smartphone **11**C is present (if "YES" in step S**94**), the smartphone **11**C returns to the process in step S**93**. If the smartphone **11**C determines that itself (smartphone **11**C) is not present in the BLE communication available area **300** (if "NO" in step S**94**), the smartphone **11**C returns to the process in step S**90**.
(240) Next, the operations of the vehicle ECU **10**C, the smartphone **11**C, and the server **14**C in the case where the server **14**C having the authentication function is provided to achieve high security will be described.
(241) FIG. **55** is a flowchart for illustrating the operation of the vehicle ECU **10**C when the server **14**C having an authentication function is provided. The operation of the vehicle ECU **10**C is the operation of the control unit **108**C, but the subject is not the control unit **108**C but the second vehicle ECU **10**C. In FIG. **55**, the vehicle ECU **10**C first determines whether the smartphone **11**C of the vehicle user **51** is in close proximity for a predetermined time or longer (step S**100**), and if

the vehicle ECU **10**C determines that the smartphone **11**C of the vehicle user **51** is not in close proximity for a predetermined time or longer (that is, if proximity is not confirmed, and if "NO" in step S**100**), the vehicle ECU **10**C determines whether a vehicle operation request is present (step S**101**). If the vehicle ECU **10**C determines that the smartphone **11**C of the vehicle user **51** is in close proximity for a predetermined time or longer (that is, if proximity is confirmed, and if "YES" in step S**100**), the vehicle ECU **10**C repeats the present process until the vehicle ECU **10**C determines that the smartphone **11**C of the vehicle user **51** is not in close proximity for a predetermined time or longer.
(242) If the vehicle ECU **10**C determines in step S**101** that the vehicle operation request is not present (if "NO" in step S**101**), the vehicle ECU **10**C repeats the present process until the vehicle ECU **10**C determines that the vehicle operation request is present. If the vehicle ECU **10**C determines that the vehicle operation request is present (if "YES" in step S**101**), the vehicle ECU **10**C detects a theft (step S**102**), and then prohibits the vehicle operation (step S**103**). Furthermore, the emergency advertising is transmitted (step S**104**).
(243) After transmitting the emergency advertising, the vehicle ECU **10**C determines whether the smartphone **11**C of the vehicle user **51** is in close proximity (step S**105**), and if the vehicle ECU **10**C determines that the smartphone **11**C is not in close proximity (if "NO" in step S**105**), the vehicle ECU **10**C returns to the process in step S**104** and continues to transmit the emergency advertising. If the vehicle ECU **10**C determines that the smartphone **11**C of the vehicle user **51** is in close proximity (if "YES" in step S**105**), the vehicle ECU **10**C receives server authentication information (step S**106**), and determines whether the server authentication is OK (step S**107**).
(244) If the vehicle ECU **10**C determines that the server authentication is not OK (that is, the server authentication is NG) (if "NO" in step S**107**), the vehicle ECU **10**C returns to the process in step S**104**. If the vehicle ECU **10**C determines that the server authentication is OK (if "YES" in step S**107**), the vehicle ECU **10**C cancels the theft detection (step S**108**), and then cancels the vehicle operation prohibition (step S**109**). Furthermore, the transmission of the emergency advertising is stopped (step S**110**). After stopping the transmission of the emergency advertising, the vehicle ECU **10**C returns to the process in step S**100**.
(245) FIG. **56** is a flowchart for illustrating the operation of the smartphone **11**C when the server **14**C having the authentication function is provided. The operation of the smartphone **11**C is an operation of the control unit **128**C, but the subject is not the control unit **128**C but the smartphone **11**C. In FIG. **56**, the smartphone **11**C is not contacted to the vehicle ECU **10**C (step S**120**), and then determines whether a predetermined time elapses (step S**121**). If the smartphone **11**C determines that the predetermined time does not elapse (if "NO" in step S**121**), the smartphone **11**C determines whether itself (smartphone **11**C) is present in the BLE communication available area **300** (step S**122**). If the smartphone **11**C determines that the smartphone **11**C is not present in the BLE communication available area **300** (if "NO" in step S**122**), the smartphone **11**C returns to the process in step S**120**, and if the smartphone **11**C determines that the smartphone **11**C is present in the BLE communication available area **300** (if "YES" in step S**122**), the smartphone **11**C performs the connection authentication with the vehicle ECU **10**C (step S**127**).
(246) On the other hand, if the smartphone **11**C determines in step S**121** that the predetermined time elapses (if "YES" in step S**121**), the smartphone **11**C performs the server authentication (step S**123**), and determines whether the server authentication is OK (step S**124**). If the smartphone **11**C determines that the server authentication is OK (if "YES" in step S**124**), the smartphone **11**C sets the server authentication information to OK (step S**125**), and if the smartphone **11**C does not determine that the server authentication is OK (if "NO" in step S**124**), the smartphone **11**C sets the server authentication information to NG (step S**126**). After performing the process in step S**125** or the process in step S**126**, the smartphone **11**C performs the process in step S**122**.
(247) After performing the connection authentication with the vehicle ECU **10**C in step S**127**, the smartphone **11**C is in the connection state with the vehicle ECU **10**C (step S**128**). Next, the

smartphone **11**C determines whether a predetermined time elapses (step S**129**), and if the smartphone **11**C determines that the predetermined time does not elapse (if "NO" in step S**129**), the smartphone **11**C determines whether the smartphone **11**C is present in the BLE communication available area **300** (step S**130**). If the smartphone **11**C determines that the smartphone **11**C is not present in the BLE communication available area **300** (if "NO" in step S**130**), the smartphone **11**C returns to the process in step S**120**, and if the smartphone **11**C determines that the smartphone **11**C is present in the BLE communication available area **300** (if "YES" in step S**130**), the smartphone **11**C performs the process in step S**128**.

(248) On the other hand, if the smartphone **11**C determines in step S**129** that the predetermined time elapses (if "YES" in step S**129**), the smartphone **11**C performs the server authentication (step S**131**), and determines whether the server authentication is OK (step S**132**). If the smartphone **11**C determines that the server authentication is OK (if "YES" in step S**132**), the smartphone **11**C sets the server authentication information to OK (step S**133**), and if the smartphone **11**C does not determine that the server authentication is OK (if "NO" in step S**132**), the smartphone **11**C sets the server authentication information to NG (step S**134**). After performing the process in step S**133** or the process in step S**134**, the smartphone **11**C performs the process in step S**130**.

(249) FIG. **57** is a flowchart for illustrating the operation of the server **14**C when the server **14**C having the authentication function is provided. Since a configuration diagram of the server **14**C is not shown, the subject will be the server **14**C as it is. However, except for having the authentication function, the server **14**C has the same configuration as the server **14** shown in FIG. **7**. In FIG. **57**, the server **14**C determines whether an authentication request is received (step S**140**), and if the server **14**C determines that the authentication request is not received (if "NO" in step S**140**), the server **14**C repeats the present process until the server **14**C determines that the authentication request is received. If the server **14**C determines that the authentication request is received (if "YES" in step S**140**), the server **14**C determines whether the user can be confirmed as the official owner (step S**141**), and if the server **14**C determines that the user can be confirmed as the official owner (if "YES" in step S**141**), the server **14**C notifies that the authentication is OK (step S**142**), and if the server **14**C determines that the user cannot be confirmed as the official owner (if "NO" in step S**141**), the server **14**C notifies that the authentication is NG (step S**143**). After performing the process in step S**142** or the process in step S**143**, the server **14**C performs the process in step S**140**.

(250) As described above, in the wireless communication system **4** according to the fourth embodiment, when the vehicle **50**C and the smartphone **11**C are in the second communication state, the vehicle **50**C can operate, and then if the vehicle **50**C and the smartphone **11**C are not in the second communication state for a predetermined time or longer, and if the main switch **504** of the vehicle **50**C detects a predetermined operation, the emergency advertising is transmitted from the vehicle **50**C, and thus, the smartphone **11**C can be used as a key that can receive the emergency information from the vehicle **50**C. By enabling the smartphone **11**C to be used as a key for the vehicle **50**C, it is possible to popularize a smartphone that can be used as a key for the vehicle **50**C.

Fifth Embodiment

(251) Next, a wireless communication system according to a fifth embodiment will be described.

(252) The wireless communication system according to the fifth embodiment sets the vehicle to an alert mode when the vehicle is powered off while the smartphone of the official owner is present in the vicinity of the vehicle ECU, measures a distance between the smartphone and the vehicle ECU by ranging technology using Bluetooth (registered trademark) communication in the alert mode, and causes the vehicle ECU to transition from the alert mode to the vehicle theft mode when the vehicle is operated without the smartphone present in the vicinity of the vehicle.

(253) Hereinafter, the wireless communication system according to the fifth embodiment will be described below with reference to FIGS. **58** to **66**.

(254) FIG. **58** is a diagram showing an aspect of using a wireless communication system **5** according to the fifth embodiment. The wireless communication system **5** according to the fifth

embodiment shown in FIG. **58** has a basic configuration of a vehicle **50**D, a vehicle ECU (wireless communication device) **10**D, and a smartphone (wireless communication terminal) **11**D. The vehicle **50**D and the smartphone **11**D belong to the vehicle user **51** who is an official owner. The vehicle ECU **10**D is mounted on the vehicle **50**D. Identification information is given to the vehicle ECU **10**D. The identification information can also be considered as identification information of the vehicle **50**D. The identification information is information uniquely determined by the vehicle ECU, and may be, for example, a MAC address.

(255) The vehicle ECU **10**D and the smartphone **11**D each have a Bluetooth communication function, and the Bluetooth communication is available between the vehicle ECU **10**D and the smartphone **11**D. The smartphone **11**D has a cellular communication function or a WiFi (registered trademark) communication function in addition to the Bluetooth communication function. The Bluetooth communication available area (BLE communication available area) **300** is an area where the Bluetooth communication is available. The proximity detection area **301** within the Bluetooth communication available area **300** is an area in the vicinity of the vehicle **50**D described above. The Bluetooth communication available area **300** has a diameter of about 1 km, and the proximity detection area **301** has a diameter of about 2 m.

(256) As described above, the vehicle ECU **10**D sets the vehicle **50**D to the alert mode when the vehicle **50**D is powered off while the smartphone **11**D of the vehicle user **51** of the vehicle **50**D is in the vicinity of the vehicle ECU **10**D. In the alert mode, a distance between the smartphone **11**D and the vehicle ECU **10**D is measured by ranging technology using the Bluetooth communication and/or UWB, and the vehicle ECU **10**D is transitioned from the alert mode to the vehicle theft mode when the vehicle **50**D is operated while the smartphone **11**D is not present in the vicinity of the vehicle **50**D.

(257) FIG. **59** is a diagram showing a case where the smartphone **11**D of the vehicle user **51** is present outside the Bluetooth communication available area **300** of the vehicle ECU **10**D. FIG. **60** is a diagram showing a case where the smartphone **11**D of the vehicle user **51** is present within the Bluetooth communication available area **300** of the vehicle ECU **10**D.

(258) Although the wireless communication system **5** according to the fifth embodiment is applied to the two-wheeled vehicle (motorcycle) **50**D, the wireless communication system **5** can of course also be applied to a four-wheeled vehicle. Hereinafter, configurations of the vehicle ECU **10**D and the smartphone **11**D of the wireless communication system **5** according to the fifth embodiment will be sequentially described.

(259) FIG. **61** is a block diagram showing a schematic configuration of a part of the vehicle **50**D and a schematic configuration of the vehicle ECU **10**D. In FIG. **61**, the same reference numerals are given to components that are common to the components of the vehicle ECU **10** (see FIG. **2**) of the wireless communication system **1** according to the first embodiment described above and the vehicle ECU **10**C (see FIG. **43**) of the wireless communication system **4** according to the fourth embodiment described above. The vehicle ECU **10**D of the wireless communication system **5** according to the fifth embodiment includes the switch detection unit **101**C, the IG authentication determination unit **102**, the BT wireless transmission and reception unit (first wireless communication circuit) **103**, the unauthorized use detection unit (unauthorized use detection circuit) **105**, the vehicle speed detection unit **106**, a distance detection unit **138**, a control unit **108**D, and the antenna **110**. The vehicle ECU **10**D has one antenna **110**, but may have a plurality thereof.

(260) The BT wireless transmission and reception unit (first wireless communication circuit) **103** performs the wireless communication conforming to the Bluetooth standard (first wireless method) as described above. The distance detection unit **138** measures a distance between the vehicle **50**D and the smartphone **11**D using Bluetooth communication and/or UWB ranging technology, and outputs the result to the control unit **108**D. A GPS reception unit (not shown) may be mounted on the vehicle **50**D (or the vehicle ECU **10**D), and the distance may be determined to the GPS

reception unit **121** (see FIG. **62**) of the smartphone **11**D.
(261) The control unit **108**D controls each unit of the device, and includes a CPU (not shown), a ROM storing a program for operating the CPU, and a RAM used in an operation of the CPU. The IG authentication determination unit **102**, the BT wireless transmission and reception unit **103**, the unauthorized use detection unit **105**, the vehicle speed detection unit **106**, and the distance detection unit **138** operate under the control of the control unit **108**D. When the acceleration sensor **1051** detects a predetermined acceleration, the unauthorized use detection unit **105** detects the unauthorized use.
(262) If the unauthorized use detection unit **105** detects the unauthorized use, and if the vehicle **50**D and the smartphone **11**D are at a predetermined distance (if the smartphone **11**D is in the Bluetooth communication available area **300**), the control unit **108**D transmits emergency information to the smartphone **11**D. The emergency information is transmitted from the BT wireless transmission and reception unit **103** of the vehicle ECU **10**D. The emergency information transmitted from the BT wireless transmission and reception unit **103** is received by the BT wireless transmission and reception unit **124** (see FIG. **62**) of the smartphone **11**D.
(263) If the unauthorized use detection unit **105** detects the unauthorized use, and if the vehicle **50**D and the smartphone **11**D are not at a predetermined distance (if the smartphone **11**D is not in the Bluetooth communication available area **300**), the control unit **108**D transmits an advertising packet of Bluetooth including the emergency information. The advertising packet of Bluetooth including the emergency information is transmitted from the BT wireless transmission and reception unit **103** of the vehicle ECU **10**D.
(264) If the unauthorized use detection unit **105** detects the unauthorized use, the control unit **108**D does not transmit the emergency information. That is, the emergency information is not transmitted from the BT wireless transmission and reception unit **103**.
(265) Here, the predetermined distance described above is a distance of the Bluetooth communication available area **300** (about 1 km in diameter), and is defined as the first distance. A distance of the proximity detection area **301** (approximately 2 m in diameter) is defined as the second distance (<first distance). If the vehicle **50**D and the smartphone **11**D are not at the second distance, and if a predetermined operation is performed by the main switch **504**, the unauthorized use detection unit **105** detects unauthorized use.
(266) A case where the BT wireless transmission and reception unit **103** of the vehicle **50**D (actually the BT wireless transmission and reception unit **103** of the vehicle ECU **10**D) and the BT wireless transmission and reception unit **124** of the smartphone **11**D (see FIG. **62**) are in a predetermined communication state is defined as the case where the vehicle **50**D and the smartphone **11**D are at a predetermined distance, and a case where the BT wireless transmission and reception unit **103** of the vehicle **50**D (actually the BT wireless transmission and reception unit **103** of the vehicle ECU **10**D) and the BT wireless transmission and reception unit **124** of the smartphone **11**D are not in a predetermined communication state is defined as the case where the vehicle **50**D and the smartphone **11**D are not at a predetermined distance.
(267) If the unauthorized use detection unit **105** of the vehicle **50**D (actually the unauthorized use detection unit **105** of the vehicle ECU **10**D) detects the unauthorized use, and if the BT wireless transmission and reception unit **103** of the vehicle **50**D (actually the BT wireless transmission and reception unit **103** of the vehicle ECU **10**D) and the BT wireless transmission and reception unit **124** of the smartphone **11**D are not in a predetermined communication state, the BT wireless transmission and reception unit **103** transmits an advertising packet of Bluetooth including the identification information of the vehicle **50**D and the emergency information.
(268) FIG. **62** is a block diagram showing a schematic configuration of the smartphone **11**D. In FIG. **44**, the same reference numerals are given to components that are common to the components of the information sharing user terminal (smartphone) **12** (see FIG. **5**) of the wireless communication system **1** according to the first embodiment described above. The smartphone **11**D

includes the GPS reception unit **121**, the BT wireless transmission and reception unit **124**, the telecom transmission and reception unit **125**, the advertising determination unit **127**, a display unit **136**, the antennas **122**, **126**, and **129**, and a control unit **128**D. The smartphone **11**D includes one antenna **126** and one antenna **129**, but may include a plurality thereof.

(269) The BT wireless transmission and reception unit (second wireless communication circuit) **124** performs the wireless communication conforming to the Bluetooth standard (first wireless method) as described above. The telecom transmission and reception unit **125** performs wireless communication conforming to a cellular communication method (second wireless method) different from the Bluetooth standard communication method (first wireless method). The display unit **136** displays information corresponding to the emergency information. The control unit **128**D controls each unit of the device, and includes a CPU (not shown), a ROM storing a program for operating the CPU, and a RAM used in an operation of the CPU. The GPS reception unit **121**, the BT wireless transmission and reception unit **124**, the telecom transmission and reception unit **125**, the advertising determination unit **127**, and the display unit **136** operate under the control of the control unit **128**D. When the control unit **128**D communicates with the vehicle ECU **10**D, the control unit **128**D transmits information for registering the control unit **128**D in the vehicle ECU **10**D.

(270) In addition to using Bluetooth, WiFi or cellular may be used for communication between the vehicle ECU **10**D and the smartphone **11**D.

(271) Next, operations of the vehicle ECU **10**D and the smartphone **11**D will be described.

(272) FIGS. **63** and **64** are sequence diagrams mainly showing the operation of the vehicle ECU **10**D. In FIG. **63**, in response to receiving the vehicle operation request from the main switch **504** of the vehicle **50**D, the vehicle ECU **10**D turns off the vehicle power of the vehicle **50**D and transitions to the alert mode. When the vehicle ECU **10**D transitions to the alert mode, the vehicle ECU **10**D performs the connection authentication and the area detection in order to connect to the smartphone **11**D of the vehicle user **51**. In this case, the smartphone **11**D is connected but not in close proximity to the vehicle **50**D (that is, the smartphone **11**D is not present within the proximity detection area **301**). In response to receiving the vehicle operation request from the main switch **504** of the vehicle **50**D in this state, the vehicle ECU **10**D detects a theft and notifies the smartphone **11**D of the vehicle user **51** of the theft.

(273) In FIG. **64**, in response to receiving the vehicle operation request from the main switch **504** of the vehicle **50**D, the vehicle ECU **10**D turns off the vehicle power of the vehicle **50**D and transitions to the alert mode. When the vehicle ECU **10**D transitions to the alert mode, the vehicle ECU **10**D performs the connection authentication and the area detection in order to connect to the smartphone **11**D of the vehicle user **51**. In this case, the smartphone **11**D cannot be connected (that is, the smartphone **11**D is not present within the BLE communication available area **300**). In response to receiving the vehicle operation request from the main switch **504** of the vehicle **50**D in this state, the vehicle ECU **10**D detects a theft and starts the transmission of the emergency advertising.

(274) FIG. **65** is a flowchart for illustrating the operation of the vehicle ECU **10**D. The operation of the vehicle ECU **10**D is an operation of the control unit **108**D, but the subject is not the control unit **108**D but the second vehicle ECU **10**D. In FIG. **65**, the vehicle ECU **10**D determines whether the vehicle operation request is received (step S**150**), and if the vehicle ECU **10**D determines that the vehicle operation request is not received (if “NO” in step S**150**), the vehicle ECU **10**D repeats the present process until the vehicle ECU **10**D determines that the vehicle operation request is received. If the vehicle ECU **10**D determines that the vehicle operation request is received (if “YES” in step S**150**), the vehicle ECU **10**D determines whether the connection is established with the smartphone **11**D of the owner (vehicle user **51**) (step S**151**). If the vehicle ECU **10**D determines that the connection is established with the smartphone **11**D of the owner (if “YES” in step S**151**), the vehicle ECU **10**D determines whether the smartphone **11**D of the owner is present in the

proximity detection area **301** (step S**152**). If the vehicle ECU **10**D determines that the smartphone **11**D of the owner is present in the proximity detection area **301** (if "YES" in step S**152**), the vehicle ECU **10**D cancels the theft detection (step S**153**), and then cancels the vehicle operation prohibition (step S**154**). Furthermore, the transmission of the emergency advertising is stopped (step S**155**). Then, the vehicle ECU **10**D returns to the process in step S**150**.
(275) On the other hand, if the vehicle ECU **10**D determines in step S**151** that the connection is not established with the smartphone **11**D of the owner (if "NO" in step S**151**), the vehicle ECU **10**D detects a theft (step S**156**), and then prohibits the vehicle operation (step S**157**). Furthermore, the transmission of the emergency advertising is started (step S**158**). Then, the vehicle ECU **10**D returns to the process in step S**150**.
(276) If the vehicle ECU **10**D determines in step S**152** that the smartphone **11**D of the owner is not present in the proximity detection area **301** (if "NO" in step S**152**), the vehicle ECU **10**D detects a theft (step S**159**), and then prohibits the vehicle operation (step S**160**). Furthermore, the smartphone **11**D of the owner is notified of the theft (step S**161**). Then, the vehicle ECU **10**D returns to the process in step S**150**.
(277) FIG. **66** is a flowchart for illustrating the operation of the smartphone **11**D. The operation of the smartphone **11**D is an operation of the control unit **128**D, but the subject is not the control unit **128**D but the smartphone **11**D. In FIG. **66**, the smartphone **11**D is first not connected to the vehicle ECU **10**D (step S**170**). Next, the smartphone **11**D determines whether itself (smartphone **11**D) is present in the BLE communication available area **300** (step S**171**). If the smartphone **11**D determines that the smartphone **11**D is not present in the BLE communication available area **300** (if "NO" in step S**171**), the smartphone **11**D returns to the process in step S**170**, and if the smartphone **11**D determines that the smartphone **11**D is present in the BLE communication available area **300** (if "YES" in step S**171**), the smartphone **11**D performs the connection authentication with the vehicle ECU **10**D (step S**172**). After performing the connection authentication with the vehicle ECU **10**D, the smartphone **11**D is in the connection state with the vehicle ECU **10**D (step S**173**). Next, the smartphone **11**D determines whether the smartphone **11**D is present in the BLE communication available area **300** (step S**174**), and if the smartphone **11**D determines that the smartphone **11**D is present in the BLE communication available area **300** (if "YES" in step S**174**), the smartphone **11**D continues the process in step S**173**. If the smartphone **11**D determines that the smartphone **11**D is not present in the BLE communication available area **300** (if "NO" in step S**174**), the smartphone **11**D returns to the process in step S**170**.
(278) As described above, in the wireless communication system **5** according to the fifth embodiment, if the unauthorized use of the vehicle **50**D is detected, and if the smartphone **11**D is at a predetermined distance from the vehicle **50**D (that is, if the smartphone **11**D is in the BLE communication available area **300**), the emergency information is transmitted from vehicle ECU **10**D of vehicle **50**D to smartphone **11**D, and if the unauthorized use of the vehicle **50**D is detected, and if the smartphone **11**D is not at a predetermined distance from the vehicle **50**D (that is, if the smartphone **11**D is not in the BLE communication available area **300**), the advertising packet of Bluetooth including the emergency information is transmitted from the vehicle ECU **10**D of the vehicle **50**D, and thus, if the smartphone **11**D is within a predetermined distance from the vehicle **50**D, the smartphone **11**D can receive the emergency information directly from the vehicle **50**D, and if the smartphone **11**D is not within a predetermined distance from the vehicle **50**D, the smartphone **11**D can indirectly receive the advertising packet of Bluetooth from the vehicle **50**D via cellular communication, WiFi, or the like.
(279) Therefore, the user **51** of the smartphone **11**D can be notified of an abnormality in the vehicle **50**D without using the telematics service in the vehicle **50**D. Since there is no need to have a communication line or a GPS function to use the telematics service, costs can be reduced. In particular, by using a communication method conforming to the Bluetooth standard for the communication between the vehicle **51**D and the smartphone **11**D, the communication between the

vehicle **51**D and the smartphone **11**D can be performed at a low cost. Accordingly, it is possible to notify an abnormality including a theft in the vehicle **50**D at a low cost and with low power consumption.
(280) If the unauthorized use detection unit **105** of the vehicle **50**D does not detect the unauthorized use, the BT wireless transmission and reception unit **103** does not transmit the emergency information, and thus, false alarms can be prevented.
(281) By using the acceleration sensor **1051** in the unauthorized use detection unit **105** of the vehicle **50**D, it is possible to detect an impact applied to the vehicle **50**D with high sensitivity.
(282) Since the information corresponding to the emergency information is displayed on the display unit **136** of the smartphone **11**D, the user **51** of the smartphone **11**D can visually confirm the information corresponding to the emergency information.
(283) Although the present invention has been described in detail with reference to a specific embodiment, it is obvious to those skilled in the art that various changes and modifications can be made without departing from the spirit and scope of the present invention.
(284) The present application also discloses a wireless communication system and a wireless communication device described in [A-1] to [A-22] below.
(285) [A-1]
(286) A wireless communication system, including: a vehicle including an unauthorized use detection circuit and a first antenna; and a wireless communication terminal including a second antenna configured to establish wireless communication conforming to a Bluetooth (registered trademark) standard with the first antenna, and a third antenna, in which if the unauthorized use detection circuit of the vehicle does not detect unauthorized use, the first antenna does not transmit an advertising packet of Bluetooth including identification information of the vehicle and emergency information, if the unauthorized use detection circuit of the vehicle detects the unauthorized use, the first antenna transmits the advertising packet of Bluetooth including the identification information of the vehicle and the emergency information, and if the wireless communication terminal receives the advertising packet of Bluetooth including the identification information of the vehicle and the emergency information via the second antenna, the wireless communication terminal transmits the identification information of the vehicle and the emergency information via the third antenna.
[A-2]
(287) The wireless communication system according to [A-1], in which the unauthorized use detection circuit includes an acceleration sensor, and if the acceleration sensor detects a predetermined acceleration, the unauthorized use is detected.
[A-3]
(288) The wireless communication system according to [A-1] or [A-2], in which the wireless communication terminal further includes a first location information detection circuit configured to detect a first latitude and longitude, and a first clock configured to acquire a first time, if the wireless communication terminal receives the advertising packet of Bluetooth including the identification information of the vehicle and the emergency information via the second antenna, the wireless communication terminal transmits the identification information of the vehicle, the emergency information, the first latitude and longitude, and the first time via the third antenna.
[A-4]
(289) The wireless communication system according to any one of [A-1] to [A-3], in which the vehicle further includes a second clock configured to acquire a second time, if the unauthorized use detection circuit of the vehicle detects the unauthorized use, the first antenna transmits the advertising packet of Bluetooth including the identification information of the vehicle, the emergency information, and the second time, and if the wireless communication terminal receives the advertising packet of Bluetooth including the identification information of the vehicle, the emergency information, and the second time via the second antenna, the wireless communication

terminal transmits the identification information of the vehicle, the emergency information, and the second time via the third antenna.
[A-5]
(290) The wireless communication system according to [A-4], in which the vehicle further includes a second location information detection circuit configured to detect a second latitude and longitude, if the unauthorized use detection circuit of the vehicle detects the unauthorized use, the first antenna transmits the advertising packet of Bluetooth including the identification information of the vehicle, the emergency information, the second time, and the second latitude and longitude, and if the wireless communication terminal receives the advertising packet of Bluetooth including the identification information of the vehicle, the emergency information, the second time, and the second latitude and longitude via the second antenna, the wireless communication terminal transmits the identification information of the vehicle, the emergency information, the second time, and the second latitude and longitude via the third antenna.
[A-6]
(291) The wireless communication system according to any one of [A-1] to [A-5], in which if the unauthorized use detection circuit of the vehicle detects the unauthorized use, the first antenna transmits the advertising packet of Bluetooth including the identification information of the vehicle and the emergency information twice or more.
[A-7]
(292) The wireless communication system according to any one of [A-1] to [A-6], in which the advertising packet of Bluetooth including the identification information of the vehicle and the emergency information is defined as a first advertising packet, the vehicle is configured to transmit the first advertising packet from the first antenna twice or more, the vehicle is configured to transmit the second advertising packet of Bluetooth, which does not include the emergency information, from the first antenna twice or more, and a first electric field strength of a radio wave with which the first advertising packet is transmitted from the first antenna is greater than a second electric field strength of a radio wave with which the second advertising packet is transmitted from the first antenna.
[A-8]
(293) The wireless communication system according to any one of [A-1] to [A-7], in which the advertising packet of Bluetooth including the identification information of the vehicle and the emergency information is defined as a first advertising packet, the vehicle is configured to transmit the first advertising packet from the first antenna twice or more, the vehicle is configured to transmit the second advertising packet of Bluetooth, which does not include the emergency information, from the first antenna twice or more, and a first transmission interval of the first advertising packet is shorter than a second transmission interval of the second advertising packet.
[A-9]
(294) The wireless communication system according to any one of [A-1] to [A-8], in which the advertising packet of Bluetooth including the identification information of the vehicle and the emergency information is defined as a first advertising packet, the vehicle is configured to transmit the first advertising packet from the first antenna twice or more, the vehicle is configured to transmit the second advertising packet of Bluetooth, which does not include the emergency information, from the first antenna twice or more, the vehicle includes a battery, and if a voltage of the battery becomes lower than a predetermined value, a first transmission interval of the first advertising packet is shorter than a second transmission interval of the second advertising packet.
[A-10]
(295) The wireless communication system according to [A-9], in which if the voltage of the battery becomes lower than the predetermined value, transmission of the second advertising packet is stopped.
[A-11]

(296) The wireless communication system according to any one of [A-1] to [A-10], further including: a server configured to wirelessly communicate with the third antenna of the wireless communication terminal, in which the server retains the identification information of the vehicle and the emergency information transmitted from the third antenna, and the held identification information and emergency information of the vehicle can be transmitted to the outside.
[A-12]
(297) A wireless communication device mountable on a vehicle, including: an unauthorized use detection circuit; and a first antenna, in which the first antenna configured to establish wireless communication conforming to a Bluetooth (registered trademark) standard with a second antenna of a wireless communication terminal, and the wireless communication terminal includes a third antenna, if the unauthorized use detection circuit of the wireless communication device does not detect unauthorized use, the first antenna does not transmit an advertising packet of Bluetooth including identification information of the wireless communication device and emergency information, if the unauthorized use detection circuit of the wireless communication device detects the unauthorized use, the first antenna transmits the advertising packet of Bluetooth including the identification information of the wireless communication device and the emergency information, and if the wireless communication terminal receives the advertising packet of Bluetooth including the identification information of the wireless communication device and the emergency information via the second antenna, the wireless communication terminal transmits the identification information of the wireless communication device and the emergency information via the third antenna.
[A-13]
(298) The wireless communication device according to [A-12], in which the unauthorized use detection circuit includes an acceleration sensor, and if the acceleration sensor detects a predetermined acceleration, the unauthorized use is detected.
[A-14]
(299) The wireless communication device according to [A-12] or [A-13], in which the wireless communication terminal further includes a first location information detection circuit configured to detect a first latitude and longitude, and a first clock configured to acquire a first time, if the wireless communication terminal receives the advertising packet of Bluetooth including the identification information of the wireless communication device and the emergency information via the second antenna, the wireless communication terminal transmits the identification information of the wireless communication device, the emergency information, the first latitude and longitude, and the first time via the third antenna.
[A-15]
(300) The wireless communication device according to any one of [A-12] to [A-14], in which the wireless communication device further includes a second clock configured to acquire a second time, if the unauthorized use detection circuit of the wireless communication device detects the unauthorized use, the first antenna transmits the advertising packet of Bluetooth including the identification information of the wireless communication device, the emergency information, and the second time, and if the wireless communication terminal receives the advertising packet of Bluetooth including the identification information of the wireless communication device, the emergency information, and the second time via the second antenna, the wireless communication terminal transmits the identification information of the wireless communication device, the emergency information, and the second time via the third antenna.
[A-16]
(301) The wireless communication device according to [A-15], further including: a second location information detection circuit configured to detect a second latitude and longitude, in which if the unauthorized use detection circuit of the wireless communication device detects the unauthorized use, the first antenna transmits the advertising packet of Bluetooth including the identification

information of the wireless communication device, the emergency information, the second time, and the second latitude and longitude, and if the wireless communication terminal receives the advertising packet of Bluetooth including the identification information of the wireless communication device, the emergency information, the second time, and the second latitude and longitude via the second antenna, the wireless communication terminal transmits the identification information of the wireless communication device, the emergency information, the second time, and the second latitude and longitude via the third antenna.
[A-17]
(302) The wireless communication device according to any one of [A-12] to [A-16], in which if the unauthorized use detection circuit of the wireless communication device detects the unauthorized use, the first antenna transmits the advertising packet of Bluetooth including the identification information of the wireless communication device and the emergency information twice or more.
[A-18]
(303) The wireless communication device according to any one of [A-12] to [A-17], in which the advertising packet of Bluetooth including the identification information of the wireless communication device and the emergency information is defined as a first advertising packet, the wireless communication device is configured to transmit the first advertising packet from the first antenna twice or more, the wireless communication device is configured to transmit the second advertising packet of Bluetooth, which does not include the emergency information, from the first antenna twice or more, and a first electric field strength of a radio wave with which the first advertising packet is transmitted from the first antenna is greater than a second electric field strength of a radio wave with which the second advertising packet is transmitted from the first antenna.
[A-19]
(304) The wireless communication device according to any one of [A-12] to [A-18], in which the advertising packet of Bluetooth including the identification information of the wireless communication device and the emergency information is defined as a first advertising packet, the wireless communication device is configured to transmit the first advertising packet from the first antenna twice or more, the wireless communication device is configured to transmit the second advertising packet of Bluetooth, which does not include the emergency information, from the first antenna twice or more, and a first transmission interval of the first advertising packet is shorter than a second transmission interval of the second advertising packet.
[A-20]
(305) The wireless communication device according to any one of [A-12] to [A-19], in which the advertising packet of Bluetooth including the identification information of the wireless communication device and the emergency information is defined as a first advertising packet, the wireless communication device is configured to transmit the first advertising packet from the first antenna twice or more, the wireless communication device is configured to transmit the second advertising packet of Bluetooth, which does not include the emergency information, from the first antenna twice or more, the vehicle includes a battery, and if a voltage of the battery becomes lower than a predetermined value, a first transmission interval of the first advertising packet is shorter than a second transmission interval of the second advertising packet.
[A-21]
(306) The wireless communication device according to [A-20], in which if the voltage of the battery becomes lower than the predetermined value, transmission of the second advertising packet is stopped.
[A-22]
(307) The wireless communication device according to any one of [A-12] to [A-21], further including: a server configured to wirelessly communicate with the third antenna of the wireless communication terminal, in which the server retains the identification information of the wireless

communication device and the emergency information transmitted from the third antenna, and the held identification information of the wireless communication device and emergency information can be transmitted to the outside.

(308) The present application also discloses a wireless communication system and a wireless communication device described in [B-1] to [B-20] below.

(309) [B-1]

(310) A wireless communication system, including: a vehicle including an operation unit and a first wireless communication circuit conforming to a first wireless method; a first wireless communication terminal which includes a second wireless communication circuit conforming to the first wireless method and available to communicate with the first wireless communication circuit; and a second wireless communication terminal which includes a third wireless communication circuit conforming to the first wireless method and available to communicate with the first wireless communication circuit, and a fourth wireless communication circuit conforming to a second wireless method different from the first wireless method, in which when the first wireless communication circuit of the vehicle and the second wireless communication circuit of the first wireless communication terminal are in a first communication state, the vehicle becomes operable, when the first wireless communication circuit of the vehicle and the third wireless communication circuit of the second wireless communication terminal are in a second communication state, the vehicle becomes operable, and after the first wireless communication circuit of the vehicle and the third wireless communication circuit of the second wireless communication terminal are in the second communication state, if the first wireless communication circuit and the third wireless communication circuit do not enter into the second communication state for a predetermined time or longer, and if the operation unit of the vehicle detects a predetermined operation, the first wireless communication circuit of the vehicle transmits a packet including emergency information.

[B-2]

(311) The wireless communication system according to [B-1], in which even when the first wireless communication circuit of the vehicle and the second wireless communication circuit of the first wireless communication terminal are in the first communication state, after the first wireless communication circuit of the vehicle and the third wireless communication circuit of the second wireless communication terminal enter into the second communication state, if the first wireless communication circuit and the third wireless communication circuit do not enter into the second communication state for the predetermined time or longer, and if the operation unit of the vehicle detects the predetermined operation, the first wireless communication circuit of the vehicle transmits the packet including the emergency information.

[B-3]

(312) The wireless communication system according to [B-1] or [B-2], in which the first wireless communication terminal does not include a wireless communication circuit conforming to the second wireless method.

[B-4]

(313) The wireless communication system according to any one of [B-1] to [B-3], in which the second wireless communication terminal is configured to stop at least the third wireless communication circuit from entering into the second communication state via the fourth wireless communication circuit.

[B-5]

(314) The wireless communication system according to any one of [B-1] to [B-4], in which the first wireless method is Bluetooth (registered trademark).

[B-6]

(315) The wireless communication system according to any one of [B-1] to [B-5], in which the second wireless method is cellular.

[B-7]
(316) The wireless communication system according to any one of [B-1] to [B-6], in which the second wireless communication terminal has at least one authentication function of password authentication, fingerprint authentication, and face authentication.
[B-8]
(317) The wireless communication system according to [B-7], in which in the authentication function of the second wireless communication terminal, if the authentication is successful, at least the third wireless communication circuit continues to be in the second communication state, and if the authentication fails, at least the third wireless communication circuit stops entering into the second communication state.
[B-9]
(318) The wireless communication system according to any one of [B-1] to [B-8], in which at least the second wireless communication terminal is registerable in the vehicle.
[B-10]
(319) The wireless communication system according to any one of [B-1] to [B-9], in which the first communication state and the second communication state are the same.
[B-11]
(320) A wireless communication device mountable on a vehicle including an operation unit, including: a first wireless communication circuit conforming to a first wireless method, in which communication with a first wireless communication terminal is available, the first wireless communication terminal which includes a second wireless communication circuit conforming to the first wireless method and available to communicate with the first wireless communication circuit, communication with a second wireless communication terminal is available, the second wireless communication terminal which includes a third wireless communication circuit conforming to the first wireless method and available to communicate with the first wireless communication circuit, and a fourth wireless communication circuit conforming to a second wireless method different from the first wireless method, when the first wireless communication circuit and the second wireless communication circuit of the first wireless communication terminal are in a first communication state, the vehicle becomes operable, when the first wireless communication circuit and the third wireless communication circuit of the second wireless communication terminal are in a second communication state, the vehicle becomes operable, and after the first wireless communication circuit and the third wireless communication circuit of the second wireless communication terminal are in the second communication state, if the first wireless communication circuit and the third wireless communication circuit do not enter into the second communication state for a predetermined time or longer, and if the operation unit of the vehicle detects a predetermined operation, the first wireless communication circuit transmits a packet including emergency information.
[B-12]
(321) The wireless communication device according to [B-11], in which even when the first wireless communication circuit and the second wireless communication circuit of the first wireless communication terminal are in the first communication state, after the first wireless communication circuit and the third wireless communication circuit of the second wireless communication terminal enter into the second communication state, if the first wireless communication circuit and the third wireless communication circuit do not enter into the second communication state for the predetermined time or longer, and if the operation unit of the vehicle detects the predetermined operation, the first wireless communication circuit transmits the packet including the emergency information.
[B-13]
(322) The wireless communication device according to [B-11] or [B-12], in which the first wireless communication terminal does not include a wireless communication circuit conforming to the

second wireless method.
[B-14]
(323) The wireless communication device according to any one of [B-11] to [B-13], in which the second wireless communication terminal is configured to stop at least the third wireless communication circuit from entering into the second communication state via the fourth wireless communication circuit.
[B-15]
(324) The wireless communication device according to any one of [B-11] to [B-14], in which the first wireless method is Bluetooth (registered trademark).
[B-16]
(325) The wireless communication device according to any one of [B-11] to [B-15], in which the second wireless method is cellular.
[B-17]
(326) The wireless communication device according to any one of [B-11] to [B-16], in which the second wireless communication terminal has at least one authentication function of password authentication, fingerprint authentication, and face authentication.
[B-18]
(327) The wireless communication device according to [B-17], in which in the authentication function of the second wireless communication terminal, if the authentication is successful, at least the third wireless communication circuit continues to be in the second communication state, and if the authentication fails, at least the third wireless communication circuit stops entering into the second communication state.
[B-19]
(328) The wireless communication device according to any one of [B-11] to [B-18], in which at least the second wireless communication terminal is registerable in the wireless communication device.
[B-20]
(329) The wireless communication device according to any one of [B-11] to [B-19], in which the first communication state and the second communication state are the same.
(330) The present application also discloses a wireless communication system and a wireless communication device described in [C-1] to [C-20] below.
(331) [C-1]
(332) A wireless communication system, including: a vehicle including an unauthorized use detection circuit and a first wireless communication circuit; and a wireless communication terminal including a second wireless communication circuit available to wirelessly communicate with the first wireless communication circuit, in which if the unauthorized use detection circuit of the vehicle detects unauthorized use, and if the vehicle and the wireless communication terminal are at a predetermined distance, the first wireless communication circuit transmits emergency information to the second wireless communication circuit, and if the unauthorized use detection circuit of the vehicle detects the unauthorized use, and if the vehicle and the wireless communication terminal are not at the predetermined distance, the first wireless communication circuit transmits an advertising packet of Bluetooth (registered trademark) including the emergency information.
[C-2]
(333) The wireless communication system according to [C-1], in which the predetermined distance is defined as a first distance, the vehicle includes an operation unit, and if the vehicle and the wireless communication terminal are not at a second distance, and if a predetermined operation is performed by the operation unit, the unauthorized use detection circuit detects the unauthorized use.
[C-3]
(334) The wireless communication system according to [C-2], in which the first distance is greater

than the second distance.
[C-4]
(335) The wireless communication system according to any one of [C-1] to [C-3], in which the first wireless communication circuit of the vehicle and the second wireless communication circuit of the wireless communication terminal conform to a Bluetooth standard.
[C-5]
(336) The wireless communication system according to [C-4], in which a distance between the vehicle and the wireless communication terminal is obtained by measurement using Bluetooth communication and/or UWB ranging technology.
[C-6]
(337) The wireless communication system according to any one of [C-1] to [C-5], in which a case where the first wireless communication circuit of the vehicle and the second wireless communication circuit of the wireless communication terminal are in a predetermined communication state is defined as the case where the vehicle and the wireless communication terminal are at the predetermined distance, and a case where the first wireless communication circuit of the vehicle and the second wireless communication circuit of the wireless communication terminal are not in the predetermined communication state is defined as the case where the vehicle and the wireless communication terminal are not at the predetermined distance.
[C-7]
(338) The wireless communication system according to any one of [C-1] to [C-6], in which if the unauthorized use detection circuit of the vehicle does not detect the unauthorized use, the first wireless communication circuit does not transmit the emergency information.
[C-8]
(339) The wireless communication system according to any one of [C-1] to [C-7], in which if the unauthorized use detection circuit of the vehicle detects the unauthorized use, and if the first wireless communication circuit of the vehicle and the second wireless communication circuit of the wireless communication terminal are not in the predetermined communication state, the first wireless communication circuit transmits an advertising packet of Bluetooth including identification information of the vehicle and the emergency information.
[C-9]
(340) The wireless communication system according to any one of [C-1] to [C-8], in which the unauthorized use detection circuit of the vehicle includes an acceleration sensor, and if the acceleration sensor detects a predetermined acceleration, the unauthorized use is detected.
[C-10]
(341) The wireless communication system according to any one of [C-1] to [C-8], in which the wireless communication terminal includes at least a display circuit, and the display circuit is configured to display information corresponding to the emergency information.
[C-11]
(342) A wireless communication device, including: an unauthorized use detection circuit; and a first wireless communication circuit, in which the wireless communication device is mountable on a vehicle and is available to communicate with a wireless communication terminal including a second wireless communication circuit available to wirelessly communicate with the first wireless communication circuit, if the unauthorized use detection circuit detects unauthorized use, and if the vehicle and the wireless communication terminal are at a predetermined distance, the first wireless communication circuit transmits emergency information to the second wireless communication circuit, and if the unauthorized use detection circuit detects the unauthorized use, and if the vehicle and the wireless communication terminal are not at the predetermined distance, the first wireless communication circuit transmits an advertising packet of Bluetooth (registered trademark) including the emergency information.
[C-12]

(343) The wireless communication device according to [C-11], in which the predetermined distance is defined as a first distance, the vehicle includes an operation unit, and if the vehicle and the wireless communication terminal are not at a second distance, and if a predetermined operation is performed by the operation unit, the unauthorized use detection circuit detects the unauthorized use.
[C-13]
(344) The wireless communication device according to [C-12], in which the first distance is greater than the second distance.
[C-14]
(345) The wireless communication device according to any one of [C-11] to [C-13], in which the first wireless communication circuit and the second wireless communication circuit of the wireless communication terminal conform to a Bluetooth standard.
[C-15]
(346) The wireless communication device according to [C-14], in which a distance between the vehicle and the wireless communication terminal is obtained by measurement using Bluetooth communication and/or UWB ranging technology.
[C-16]
(347) The wireless communication device according to any one of [C-11] to [C-15], in which a case where the first wireless communication circuit and the second wireless communication circuit of the wireless communication terminal are in a predetermined communication state is defined as the case where the vehicle and the wireless communication terminal are at the predetermined distance, and a case where the first wireless communication circuit and the second wireless communication circuit of the wireless communication terminal are not in the predetermined communication state is defined as the case where the vehicle and the wireless communication terminal are not at the predetermined distance.
[C-17]
(348) The wireless communication device according to any one of [C-11] to [C-16], in which if the unauthorized use detection circuit does not detect the unauthorized use, the first wireless communication circuit does not transmit the emergency information.
[C-18]
(349) The wireless communication device according to any one of [C-11] to [C-17], in which if the unauthorized use detection circuit detects the unauthorized use, and if the first wireless communication circuit and the second wireless communication circuit of the wireless communication terminal are not in the predetermined communication state, the first wireless communication circuit transmits an advertising packet of Bluetooth including identification information of the vehicle and the emergency information.
[C-19]
(350) The wireless communication device according to any one of [C-11] to [C-18], in which the unauthorized use detection circuit includes an acceleration sensor, and if the acceleration sensor detects a predetermined acceleration, the unauthorized use is detected.
[C-20]
(351) The wireless communication device according to any one of [C-11] to [C-18], in which the wireless communication terminal includes at least a display circuit, and the display circuit is configured to display information corresponding to the emergency information.
(352) The present application also discloses a wireless communication system and a wireless communication device described in [D-1] to [D-20] below.
(353) [D-1]
(354) A wireless communication system, including: a first vehicle including an unauthorized use detection circuit and a first antenna; and a second vehicle including a second antenna, in which if the unauthorized use detection circuit of the first vehicle detects unauthorized use, the first antenna

transmits an advertising packet of Bluetooth (registered trademark) including identification information of the first vehicle and emergency information, and if the second vehicle receives the advertising packet of Bluetooth including the identification information of the first vehicle and the emergency information via the second antenna, the second antenna transmits the predetermined packet of Bluetooth including the identification information of the first vehicle and the emergency information.

[D-2]

(355) The wireless communication system according to [D-1], in which the advertising packet is defined as a first advertising packet, and the predetermined packet is defined as a second advertising packet.

[D-3]

(356) The wireless communication system according to [D-1] or [D-2], further including: a wireless communication terminal including a fourth antenna and a third antenna capable of wireless communication conforming to a Bluetooth standard with the second antenna of the second vehicle, in which if the wireless communication terminal receives the predetermined packet of Bluetooth including the identification information of the first vehicle and the emergency information via the third antenna, the wireless communication terminal transmits the identification information of the first vehicle and the emergency information via the fourth antenna.

[D-4]

(357) The wireless communication system according to any one of [D-1] to [D-3], in which the second vehicle includes a memory circuit configured to store at least the identification information of the first vehicle.

[D-5]

(358) The wireless communication system according to any one of [D-1] to [D-4], in which the second antenna of the second vehicle transmits the predetermined packet of Bluetooth including the identification information of the first vehicle and the emergency information twice or more.

[D-6]

(359) The wireless communication system according to any one of [D-1] to [D-5], in which the unauthorized use detection circuit of the first vehicle includes an acceleration sensor, and if the acceleration sensor detects a predetermined acceleration, the unauthorized use is detected.

[D-7]

(360) The wireless communication system according to [D-3], in which the fourth antenna of the wireless communication terminal can be used for cellular communication.

[D-8]

(361) The wireless communication system according to [D-3] or [D-7], in which the wireless communication terminal further includes a first location information detection circuit configured to detect a first latitude and longitude, and a first clock configured to acquire a first time, if the wireless communication terminal receives the predetermined packet of Bluetooth including the identification information of the first vehicle and the emergency information via the third antenna, the wireless communication terminal transmits the identification information of the first vehicle, the emergency information, the first latitude and longitude, and the first time via the fourth antenna.

[D-9]

(362) The wireless communication system according to [D-3], [D-7], or [D-8], in which the second vehicle further includes a second clock configured to acquire a second time, if the second vehicle receives the advertising packet of Bluetooth including the identification information of the first vehicle and the emergency information via the second antenna, the second antenna transmits the predetermined packet of Bluetooth including the identification information of the first vehicle, the emergency information, and the second time, and if the wireless communication terminal receives the predetermined packet of Bluetooth including the identification information of the first vehicle, the emergency information, and the second time via the third antenna, the wireless communication

terminal transmits the identification information of the first vehicle, the emergency information, and the second time via the fourth antenna.

[D-10]

(363) The wireless communication system according to [D-9], in which the second vehicle further includes a second location information detection circuit configured to acquire a second latitude and longitude, if the second vehicle receives the advertising packet of Bluetooth including the identification information of the first vehicle and the emergency information via the second antenna, the second antenna transmits the predetermined packet of Bluetooth including the identification information of the first vehicle, the emergency information, the second time, and the second latitude and longitude, and if the wireless communication terminal receives the predetermined packet of Bluetooth including the identification information of the first vehicle, the emergency information, the second time, and the second latitude and longitude via the third antenna, the wireless communication terminal transmits the identification information of the first vehicle, the emergency information, the second time, and the second latitude and longitude via the fourth antenna.

[D-11]

(364) A wireless communication device mountable on a second vehicle, in which a wireless communication system includes a first vehicle including an unauthorized use detection circuit and a first antenna, and the second vehicle including a second antenna, if the unauthorized use detection circuit of the first vehicle detects unauthorized use, the first antenna transmits an advertising packet of Bluetooth (registered trademark) including identification information of the first vehicle and emergency information, and if the wireless communication device of the second vehicle receives the advertising packet of Bluetooth including the identification information of the first vehicle and the emergency information via the second antenna, the second antenna transmits the predetermined packet of Bluetooth including the identification information of the first vehicle and the emergency information.

[D-12]

(365) The wireless communication device according to [D-11], in which the advertising packet is defined as a first advertising packet, and the predetermined packet is defined as a second advertising packet.

[D-13]

(366) The wireless communication device according to [D-11] or [D-12], in which communication with a wireless communication terminal via the second antenna is available, the wireless communication terminal including a fourth antenna and a third antenna capable of wireless communication conforming to a Bluetooth standard with the second antenna of the second vehicle, and if the wireless communication terminal receives the predetermined packet of Bluetooth including the identification information of the first vehicle and the emergency information via the third antenna, the wireless communication terminal transmits the identification information of the first vehicle and the emergency information via the fourth antenna.

[D-14]

(367) The wireless communication device according to any one of [D-11] to [D-13], further including: a memory circuit configured to store at least the identification information of the first vehicle.

[D-15]

(368) The wireless communication device according to any one of [D-11] to [D-14], in which the second antenna of the second vehicle transmits the predetermined packet of Bluetooth including the identification information of the first vehicle and the emergency information twice or more.

[D-16]

(369) The wireless communication device according to any one of [D-11] to [D-15], in which the unauthorized use detection circuit of the first vehicle includes an acceleration sensor, and if the

acceleration sensor detects a predetermined acceleration, the unauthorized use is detected.
[D-17]
(370) The wireless communication device according to [D-13], in which the fourth antenna of the wireless communication terminal can be used for cellular communication.
[D-18]
(371) The wireless communication device according to [D-13] or [D-17], in which the wireless communication terminal further includes a first location information detection circuit configured to detect a first latitude and longitude, and a first clock configured to acquire a first time, if the wireless communication terminal receives the predetermined packet of Bluetooth including the identification information of the first vehicle and the emergency information via the third antenna, the wireless communication terminal transmits the identification information of the first vehicle, the emergency information, the first latitude and longitude, and the first time via the fourth antenna.
[D-19]
(372) The wireless communication device according to [D-13], [D-17], or [D-18], further including: a second clock configured to acquire a second time, in which if the wireless communication device receives the advertising packet of Bluetooth including the identification information of the first vehicle and the emergency information via the second antenna, the second antenna transmits the predetermined packet of Bluetooth including the identification information of the first vehicle, the emergency information, and the second time, and if the wireless communication terminal receives the predetermined packet of Bluetooth including the identification information of the first vehicle, the emergency information, and the second time via the third antenna, the wireless communication terminal transmits the identification information, the emergency information, and the second time of the first vehicle via the fourth antenna.
[D-20]
(373) The wireless communication device according to [D-19], further including: a second location information detection circuit configured to acquire a second latitude and longitude, in which if the wireless communication device receives the advertising packet of Bluetooth including the identification information of the first vehicle and the emergency information via the second antenna, the second antenna transmits the predetermined packet of Bluetooth including the identification information of the first vehicle, the emergency information, the second time, and the second latitude and longitude, and if the wireless communication terminal receives the predetermined packet of Bluetooth including the identification information of the first vehicle, the emergency information, the second time, and the second latitude and longitude via the third antenna, the wireless communication terminal transmits the identification information of the first vehicle, the emergency information, the second time, and the second latitude and longitude via the fourth antenna.
INDUSTRIAL APPLICABILITY
(374) The wireless communication system according to the present disclosure is useful for low-cost vehicles such as a scooter.
REFERENCE SIGNS LIST
(375) **1** to **5** wireless communication system **10**, **10**C, **10**D vehicle ECU **10**A, **10**F first vehicle ECU **10**B, **10**E second vehicle ECU **11** vehicle user terminal **11**C, **11**D, **11**E smartphone **12**, **12**B, **12**C, **12**D information sharing user terminal **13** bicycle parking lot Bluetooth unit **14**, **14**B, **14**C, **14**D server **15** telecom base station **16**, **16**E electronic key **50**, **50**C, **50**D vehicle **50**A, **50**F first vehicle **50**B, **50**E second vehicle **51** vehicle user (official owner) **52** information sharing user **53** cloud **101**, **101**C switch detection unit **102** IG authentication determination unit **103**, **124**, **131**, **161** BT wireless transmission and reception unit **104**, **121** GPS reception unit **105** unauthorized use detection unit **106** vehicle speed detection unit **107**, **123** clock **108**, **108**A, **108**B, **108**C, **108**D, **108**E, **108**F, **128**, **128**B, **128**C, **128**D, **128**E, **135**, **143**, **143**B, **143**C, **164** control unit **109** memory **110**, **111**, **122**, **126**, **129**, **132**, **165** antenna **112** advertising detection unit **125** telecom transmission

and reception unit **127**, **134** advertising determination unit **133**, **141** network communication unit **136** display unit **137** authentication unit **138** distance detection unit **142** notification information determination unit **162** key switch **163** LED **298** communication range of normal advertising **299** communication range of emergency advertising **300** Bluetooth communication available area (BLE communication available area) **301** proximity detection area **501** glove box switch **502** IG KEY switch **503** vehicle speed management ECU **504** main switch **1051** acceleration sensor

Claims

1. A wireless communication system, comprising: a vehicle including an operation unit and a first wireless communication circuit conforming to a first wireless method; a first wireless communication terminal including a second wireless communication circuit conforming to the first wireless method and available to communicate with the first wireless communication circuit; and a second wireless communication terminal including a third wireless communication circuit conforming to the first wireless method and available to communicate with the first wireless communication circuit, and a fourth wireless communication circuit conforming to a second wireless method different from the first wireless method, wherein when the first wireless communication circuit of the vehicle and the second wireless communication circuit of the first wireless communication terminal are in a first communication state, the vehicle becomes operable, when the first wireless communication circuit of the vehicle and the third wireless communication circuit of the second wireless communication terminal are in a second communication state, the vehicle becomes operable, and after the first wireless communication circuit of the vehicle and the third wireless communication circuit of the second wireless communication terminal are in the second communication state, if the first wireless communication circuit and the third wireless communication circuit do not enter into the second communication state for a predetermined time or longer, and if the operation unit of the vehicle detects a predetermined operation, the first wireless communication circuit of the vehicle transmits a packet including emergency information.
2. The wireless communication system according to claim 1, wherein even when the first wireless communication circuit of the vehicle and the second wireless communication circuit of the first wireless communication terminal are in the first communication state, after the first wireless communication circuit of the vehicle and the third wireless communication circuit of the second wireless communication terminal enter into the second communication state, if the first wireless communication circuit and the third wireless communication circuit do not enter into the second communication state for the predetermined time or longer, and if the operation unit of the vehicle detects the predetermined operation, the first wireless communication circuit of the vehicle transmits the packet including the emergency information.
3. The wireless communication system according to claim 1, wherein the first wireless communication terminal does not include a wireless communication circuit conforming to the second wireless method.
4. The wireless communication system according to claim 1, wherein the second wireless communication terminal is configured to stop at least the third wireless communication circuit from entering into the second communication state via the fourth wireless communication circuit.
5. The wireless communication system according to claim 1, wherein the first wireless method is Bluetooth (registered trademark).
6. The wireless communication system according to claim 1, wherein the second wireless method is cellular.
7. The wireless communication system according to claim 1, wherein the second wireless communication terminal has at least one authentication function of password authentication, fingerprint authentication, and face authentication.
8. The wireless communication system according to claim 7, wherein in the authentication function

of the second wireless communication terminal, if the authentication is successful, at least the third wireless communication circuit continues to be in the second communication state, and if the authentication fails, at least the third wireless communication circuit stops entering into the second communication state.

9. The wireless communication system according to claim 1, wherein at least the second wireless communication terminal is registerable in the vehicle.

10. The wireless communication system according to claim 1, wherein the first communication state and the second communication state are the same.

11. A wireless communication device mountable on a vehicle including an operation unit, comprising: a first wireless communication circuit conforming to a first wireless method, wherein the wireless communication device is able to be communicated with a first wireless communication terminal, the first wireless communication terminal including a second wireless communication circuit conforming to the first wireless method and available to communicate with the first wireless communication circuit, the wireless communication device is able to be communicated with a second wireless communication terminal, the second wireless communication terminal including a third wireless communication circuit conforming to the first wireless method and available to communicate with the first wireless communication circuit, and a fourth wireless communication circuit conforming to a second wireless method different from the first wireless method, when the first wireless communication circuit and the second wireless communication circuit of the first wireless communication terminal are in a first communication state, the vehicle becomes operable, when the first wireless communication circuit and the third wireless communication circuit of the second wireless communication terminal are in a second communication state, the vehicle becomes operable, and after the first wireless communication circuit and the third wireless communication circuit of the second wireless communication terminal are in the second communication state, if the first wireless communication circuit and the third wireless communication circuit do not enter into the second communication state for a predetermined time or longer, and if the operation unit of the vehicle detects a predetermined operation, the first wireless communication circuit transmits a packet including emergency information.

12. The wireless communication device according to claim 11, wherein even when the first wireless communication circuit and the second wireless communication circuit of the first wireless communication terminal are in the first communication state, after the first wireless communication circuit and the third wireless communication circuit of the second wireless communication terminal enter into the second communication state, if the first wireless communication circuit and the third wireless communication circuit do not enter into the second communication state for the predetermined time or longer, and if the operation unit of the vehicle detects the predetermined operation, the first wireless communication circuit transmits the packet including the emergency information.

13. The wireless communication device according to claim 11, wherein the first wireless communication terminal does not include a wireless communication circuit conforming to the second wireless method.

14. The wireless communication device according to claim 11, wherein the second wireless communication terminal is configured to stop at least the third wireless communication circuit from entering into the second communication state via the fourth wireless communication circuit.

15. The wireless communication device according to claim 11, wherein the first wireless method is Bluetooth (registered trademark).

16. The wireless communication device according to claim 11, wherein the second wireless method is cellular.

17. The wireless communication device according to claim 11, wherein the second wireless communication terminal has at least one authentication function of password authentication, fingerprint authentication, and face authentication.

18. The wireless communication device according to claim 17, wherein in the authentication function of the second wireless communication terminal, if the authentication is successful, at least the third wireless communication circuit continues to be in the second communication state, and if the authentication fails, at least the third wireless communication circuit stops entering into the second communication state.
19. The wireless communication device according to claim 11, wherein at least the second wireless communication terminal is registerable in the wireless communication device.
20. The wireless communication device according to claim 11, wherein the first communication state and the second communication state are the same.